

Shot point notebook

Universal Trench · FITASC Universal 1 · 25 clays

What you have in your hands

A complete round, clay by clay, exactly as the shooter would see it from the station. For each clay the whole trajectory is drawn, with five moments marked on it, spread across the window in which the clay is still within reach, plus one that is already late.

At each moment it tells you where the clay is and how much lead you need to break it, measured as the eye sees it. It is meant for planning the round before you go to the ground: what you shoot first, how much you lead, and what happens to you if you are late.

How to read each sheet

- The large figure is the lead as the eye perceives it, perpendicular to the line of sight. Below it also appears in clay widths and in degrees.
- The real three-dimensional separation is in small print: it includes the depth, which the eye does not see, and that is why it is a much larger number.
- Every sheet shares the same scale, so leads are comparable from one clay to another. The header says how much the view has been turned and with what framing.
- The colour of each moment warns you about distance: within reach, at the limit, or already beyond useful range.

What it is not

- It is not a field measurement: these are figures from a teaching model, with its simplifications. The last page explains this in detail.
- It does not certify an installation or a trajectory. In competition the authorised officials always decide.
- It does not replace an instructor, the rules of the ground, or the legal requirements.

Generated with these settings

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral

Every lead in this notebook is valid only for that combination. With another cartridge, another rib, another choke or with wind, the figures change and this notebook stops being useful: you have to generate a new one.

You do that from the simulator, changing whatever you want and pressing «Generate notebook» again. On each sheet, the settings that are not the start-up ones appear highlighted in colour.

The project

Plato is a free simulator and book for learning clay target shooting, written by Roberto Canales Mora. The book explains the ground, safety, the disciplines and the geometry of lead; the simulator lets you repeat the same clay changing one single variable.

Project, contact and repository on GitHub:

www.robertocanales.com/proyectos#plato
rcanalescoder.github.io/Plato

What is on the following pages

- An index of the round: a table with station, machine, angle, height, fall and lead for each clay, and a plan view with all the trajectories together.
- Una ficha por plato, 25 en total, en el orden en que los verías durante la tirada.
- The notice of use and the licence, at the end. If you print only part of it, print that sheet too.

Notice: the figures in this notebook are an estimate from a teaching model, not a field measurement. Read the notice of use and the licence on the last page before taking it to the ground.

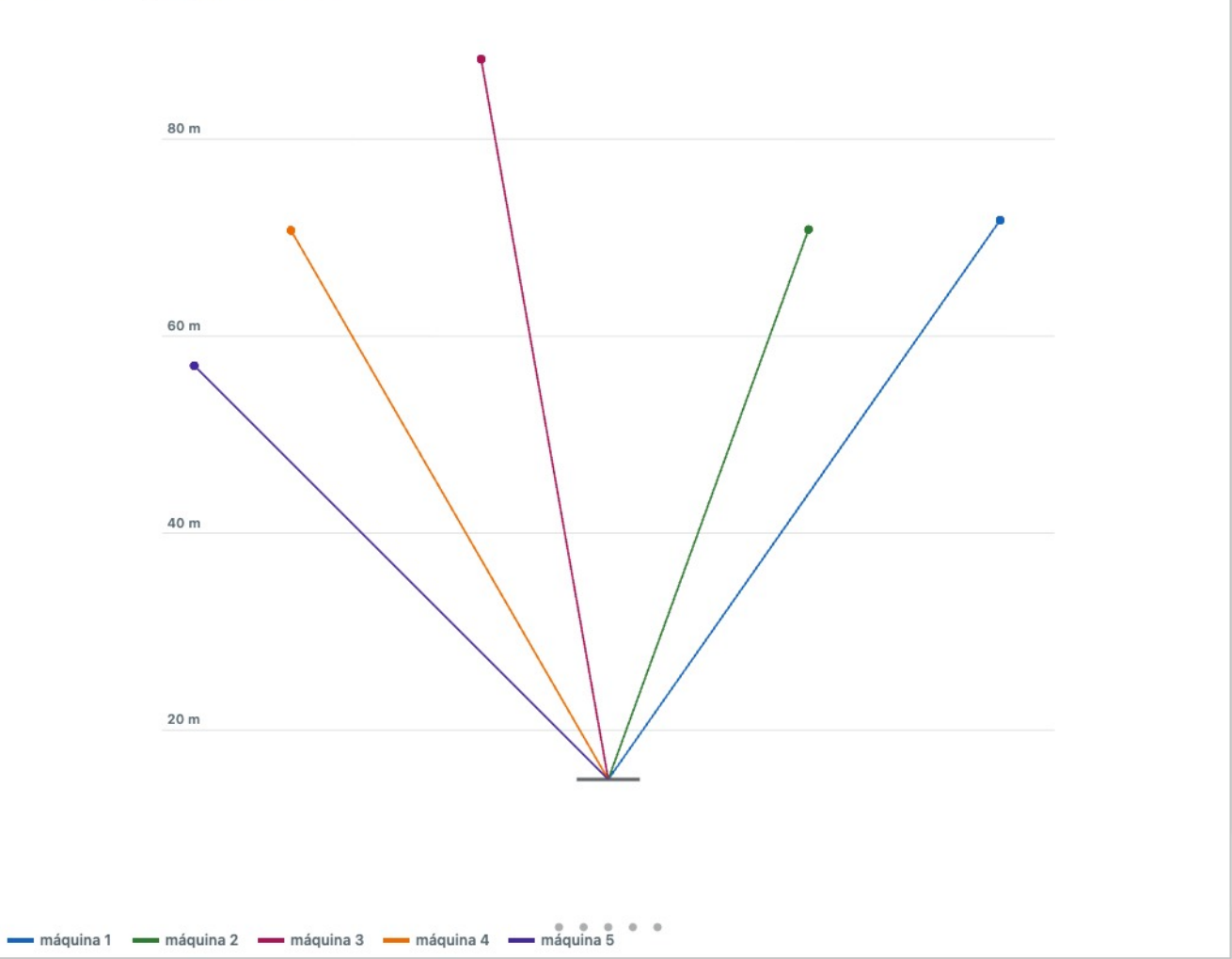
Shot point notebook

Universal Trench · FITASC Universal 1 · 25 clays

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral

#	station	mach.	angle	h. 10 m	fall	lead in the good window
1	1	1	+35°	1.5 m	70 m	71 cm · 6.5 platos · 0.78 s
2	1	2	+20°	2.0 m	60 m	5 cm · 0.5 platos · 0.97 s
3	1	3	−10°	1.5 m	75 m	124 cm · 11.3 platos · 0.77 s
4	1	4	−30°	3.0 m	65 m	113 cm · 10.2 platos · 1.25 s
5	1	5	−45°	1.5 m	60 m	193 cm · 17.5 platos · 1.00 s
6	2	1	+35°	1.5 m	70 m	102 cm · 9.3 platos · 0.81 s
7	2	2	+20°	2.0 m	60 m	32 cm · 2.9 platos · 0.99 s
8	2	3	−10°	1.5 m	75 m	84 cm · 7.6 platos · 0.77 s
9	2	4	−30°	3.0 m	65 m	92 cm · 8.4 platos · 1.23 s
10	2	5	−45°	1.5 m	60 m	170 cm · 15.4 platos · 0.97 s
11	3	1	+35°	1.5 m	70 m	133 cm · 12.1 platos · 0.83 s
12	3	2	+20°	2.0 m	60 m	60 cm · 5.4 platos · 1.01 s
13	3	3	−10°	1.5 m	75 m	44 cm · 4.0 platos · 0.77 s
14	3	4	−30°	3.0 m	65 m	72 cm · 6.5 platos · 1.20 s
15	3	5	−45°	1.5 m	60 m	146 cm · 13.3 platos · 0.94 s
16	4	1	+35°	1.5 m	70 m	165 cm · 15.0 platos · 0.86 s
17	4	2	+20°	2.0 m	60 m	87 cm · 7.9 platos · 1.03 s
18	4	3	−10°	1.5 m	75 m	11 cm · 1.0 platos · 0.76 s
19	4	4	−30°	3.0 m	65 m	51 cm · 4.7 platos · 1.17 s
20	4	5	−45°	1.5 m	60 m	122 cm · 11.1 platos · 0.90 s
21	5	1	+35°	1.5 m	70 m	195 cm · 17.7 platos · 0.88 s
22	5	2	+20°	2.0 m	60 m	113 cm · 10.3 platos · 1.04 s
23	5	3	−10°	1.5 m	75 m	40 cm · 3.7 platos · 0.75 s
24	5	4	−30°	3.0 m	65 m	30 cm · 2.8 platos · 1.13 s
25	5	5	−45°	1.5 m	60 m	99 cm · 9.0 platos · 0.87 s

Plan view · every clay in the scheme



The table uses the «good window» moment, the comfortable shot; each sheet details the other moments.

The lead is the one the shooter sees: the apparent separation between clay and aiming point, measured perpendicular to the line of sight.

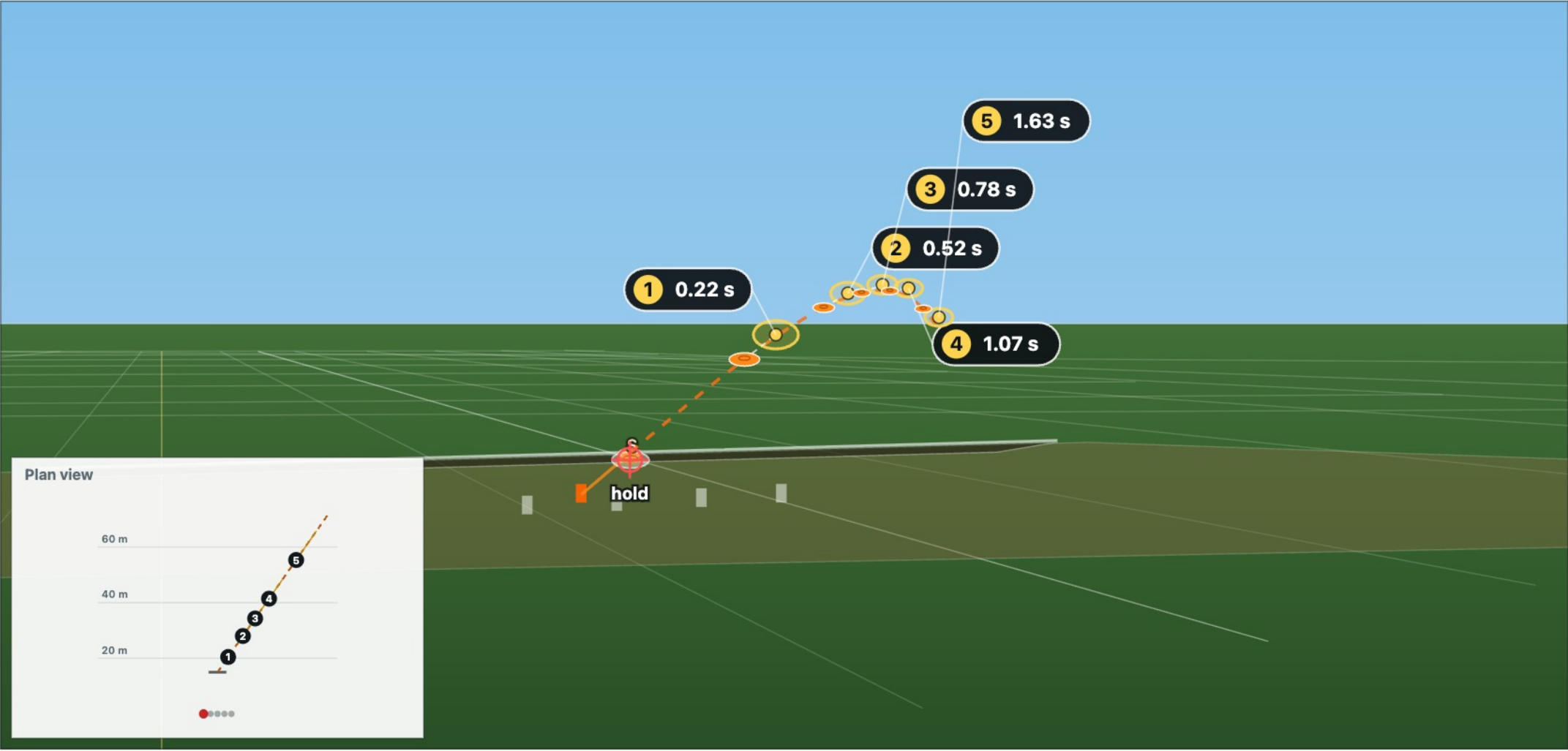
Red: the gun's hold point. Orange: the clay. Yellow: where to put the aiming point. Yellow ellipse: estimated pattern of the first shot.

Station 1 · Machine 1

angle +35° · height 1.5 m at 10 m · fall 70 m · 110 km/h · elevation 8.9°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned +7° horizontally and +4° vertically



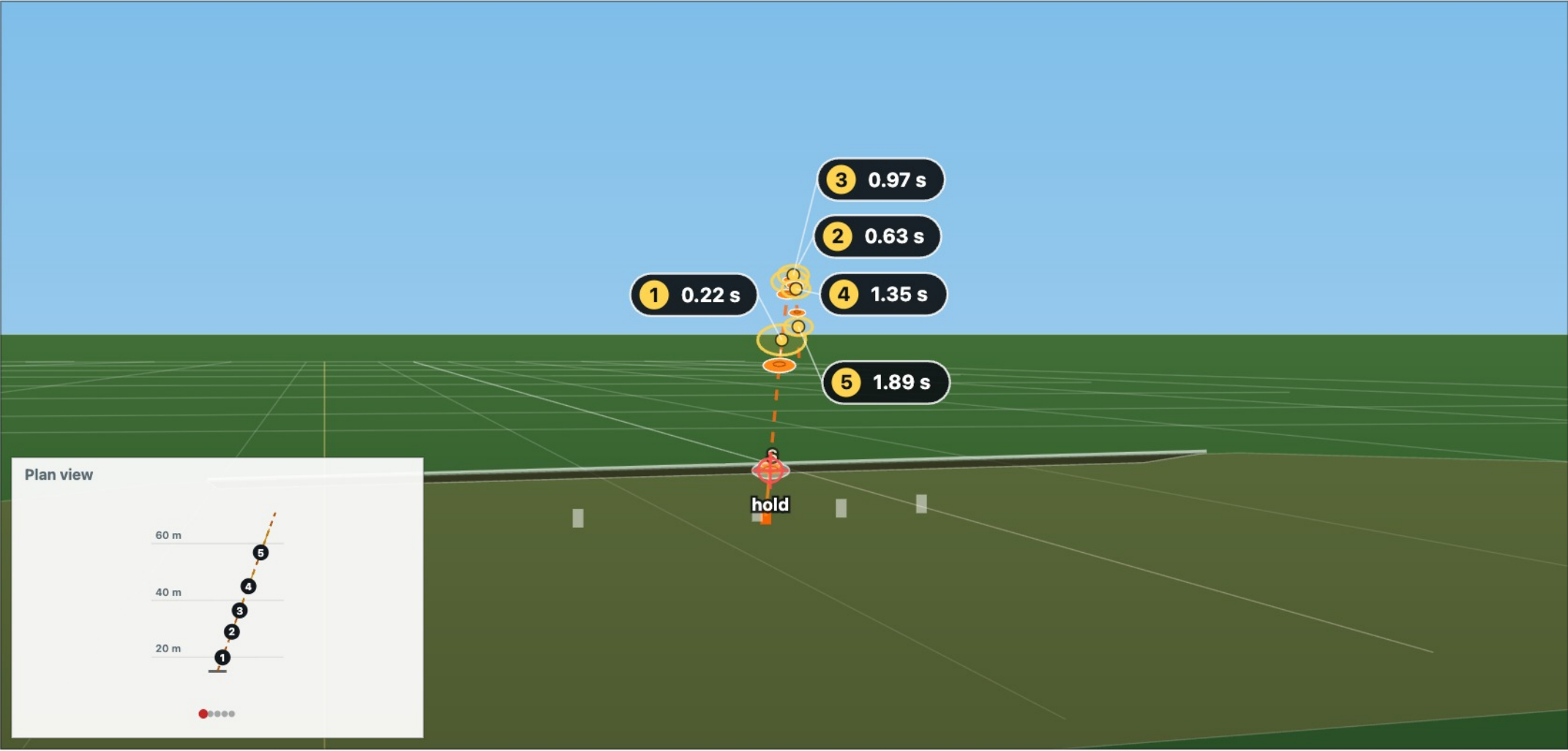
1 Release 0.22 s after release 73 cm ≈ 6.6 clays 59 cm right · 43 cm up +1.52° lateral · +1.10° vertical clay at 22 m · height 1.1 m · real separation 3.0 m first glimpse · within reach	2 Early 0.52 s after release 71 cm ≈ 6.5 clays 65 cm right · 29 cm up +1.19° lateral · +0.52° vertical clay at 31 m · height 2.1 m · real separation 4.5 m quick shot · within reach	3 Good window 0.78 s after release 71 cm ≈ 6.5 clays 71 cm right · 9 cm up +1.04° lateral · +0.13° vertical clay at 39 m · height 2.6 m · real separation 6.1 m the comfortable shot · at the limit	4 At the limit 1.07 s after release 82 cm ≈ 7.5 clays 78 cm right · 26 cm down +0.94° lateral · -0.31° vertical clay at 48 m · height 2.9 m · real separation 8.2 m last useful shot · at the limit	5 You are late 1.63 s after release 160 cm ≈ 14.5 clays 89 cm right · 133 cm down +0.79° lateral · -1.18° vertical clay at 65 m · height 2.4 m · real separation 12.8 m no longer reaching well · beyond useful range
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Station 1 · Machine 2

angle +20° · height 2.0 m at 10 m · fall 60 m · 87 km/h · elevation 12.5°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned +1° horizontally and +4° vertically



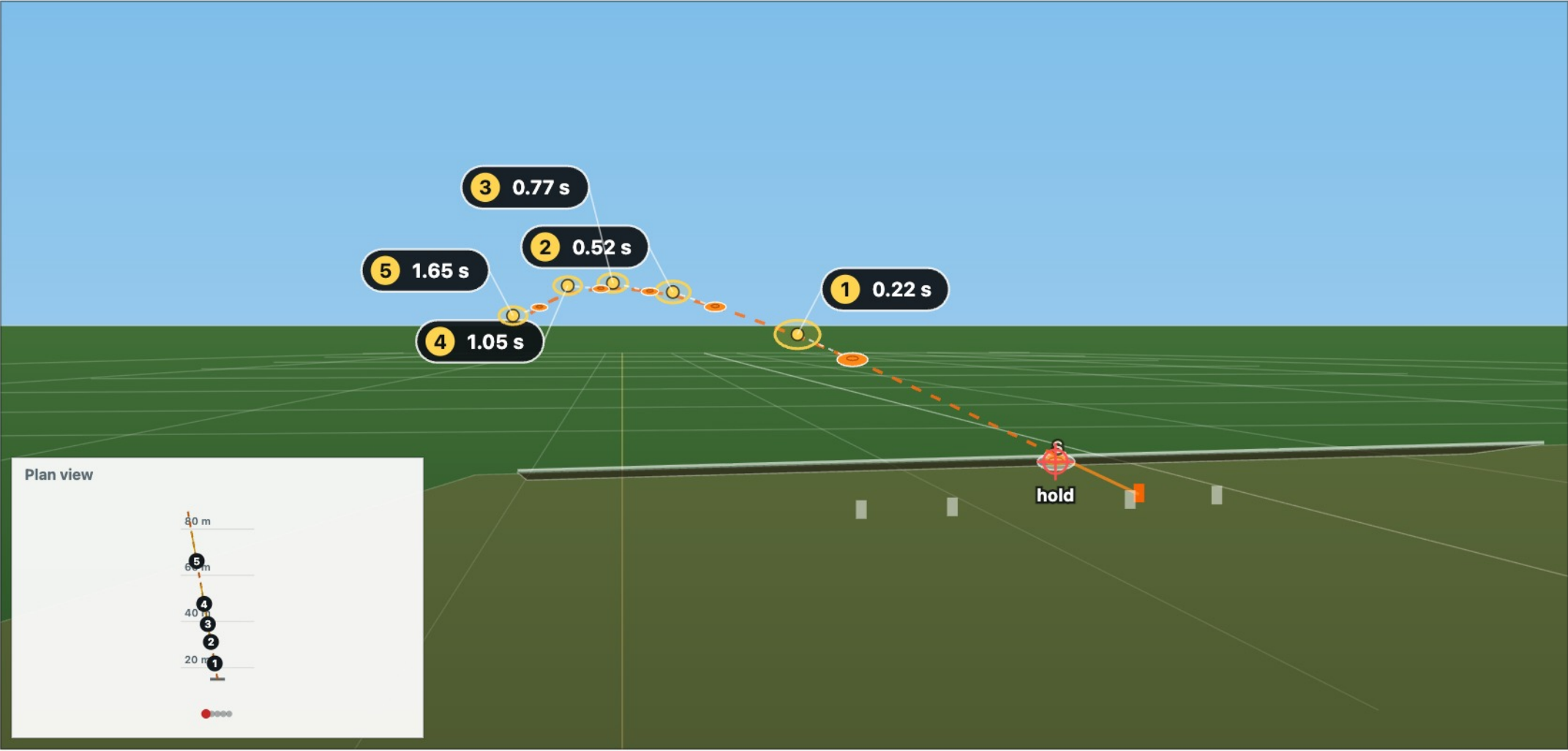
1 Release 0.22 s after release 41 cm ≈ 3.7 clays 4 cm right · 41 cm up +0.12° lateral · +1.11° vertical clay at 21 m · height 1.2 m · real separation 2.1 m first glimpse · within reach	2 Early 0.63 s after release 23 cm ≈ 2.1 clays 5 cm right · 23 cm up +0.09° lateral · +0.43° vertical clay at 31 m · height 2.6 m · real separation 3.3 m quick shot · within reach	3 Good window 0.97 s after release 5 cm ≈ 0.5 clays 5 cm right · 2 cm down +0.07° lateral · -0.03° vertical clay at 39 m · height 3.3 m · real separation 4.5 m the comfortable shot · at the limit	4 At the limit 1.35 s after release 45 cm ≈ 4.1 clays 6 cm right · 45 cm down +0.07° lateral · -0.54° vertical clay at 48 m · height 3.4 m · real separation 6.1 m last useful shot · at the limit	5 You are late 1.89 s after release 139 cm ≈ 12.6 clays 6 cm right · 139 cm down +0.06° lateral · -1.31° vertical clay at 60 m · height 2.7 m · real separation 8.9 m no longer reaching well · beyond useful range
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Station 1 · Machine 3

angle -10° · height 1.5 m at 10 m · fall 75 m · 115 km/h · elevation 8.8°

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cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned -11° horizontally and +4° vertically



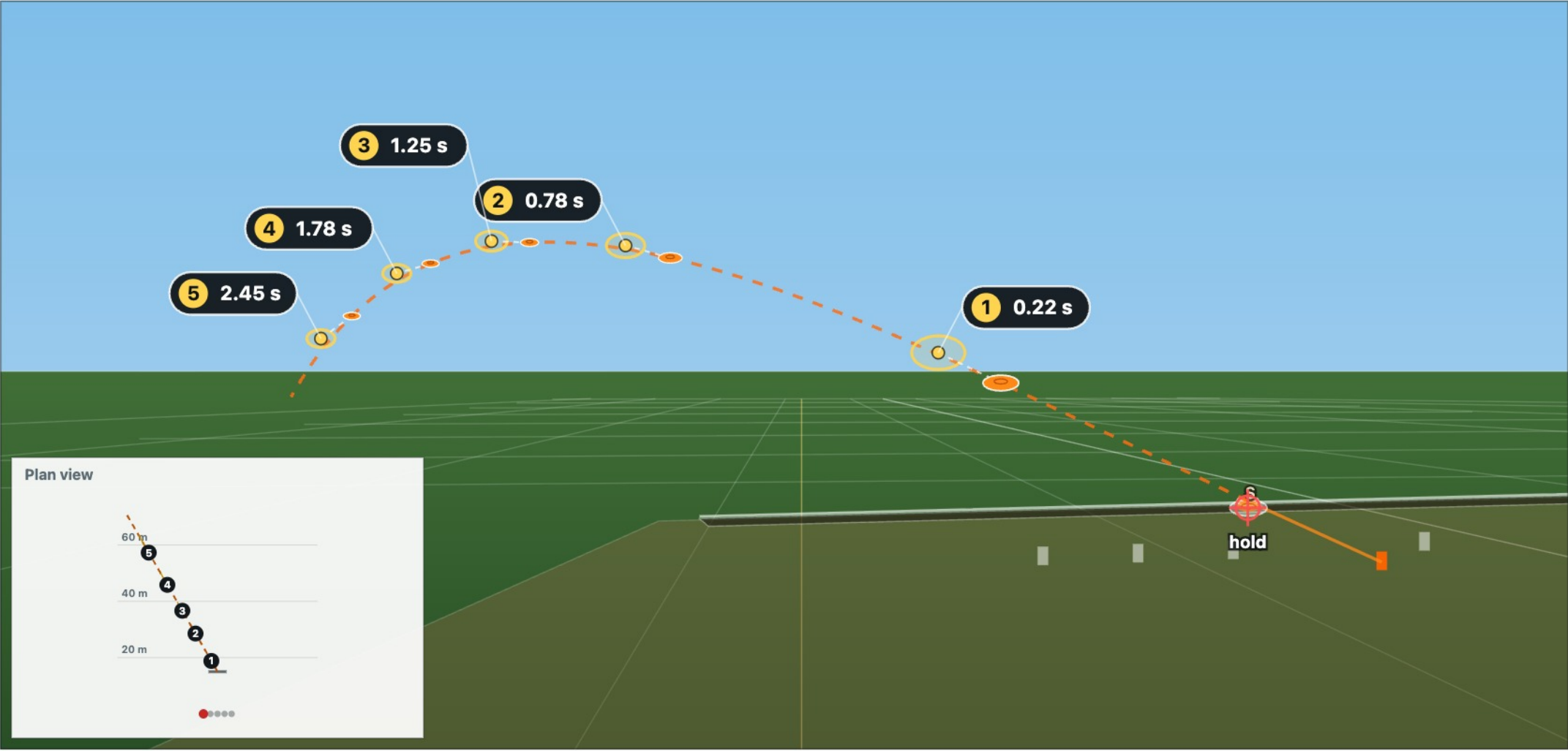
1 Release 0.22 s after release 112 cm ≈ 10.2 clays 103 cm left · 44 cm up -2.66° lateral · +1.13° vertical clay at 22 m · height 1.1 m · real separation 3.1 m first glimpse · within reach	2 Early 0.52 s after release 117 cm ≈ 10.7 clays 113 cm left · 30 cm up -2.08° lateral · +0.55° vertical clay at 31 m · height 2.1 m · real separation 4.7 m quick shot · within reach	3 Good window 0.77 s after release 124 cm ≈ 11.3 clays 123 cm left · 10 cm up -1.82° lateral · +0.15° vertical clay at 39 m · height 2.7 m · real separation 6.4 m the comfortable shot · at the limit	4 At the limit 1.05 s after release 139 cm ≈ 12.6 clays 137 cm left · 24 cm down -1.64° lateral · -0.29° vertical clay at 48 m · height 3.0 m · real separation 8.6 m last useful shot · at the limit	5 You are late 1.65 s after release 202 cm ≈ 18.3 clays 150 cm left · 134 cm down -1.30° lateral · -1.16° vertical clay at 66 m · height 2.6 m · real separation 13.3 m no longer reaching well · beyond useful range
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Station 1 · Machine 4

angle -30° · height 3.0 m at 10 m · fall 65 m · 76 km/h · elevation 18.6°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned -19° horizontally and +6° vertically



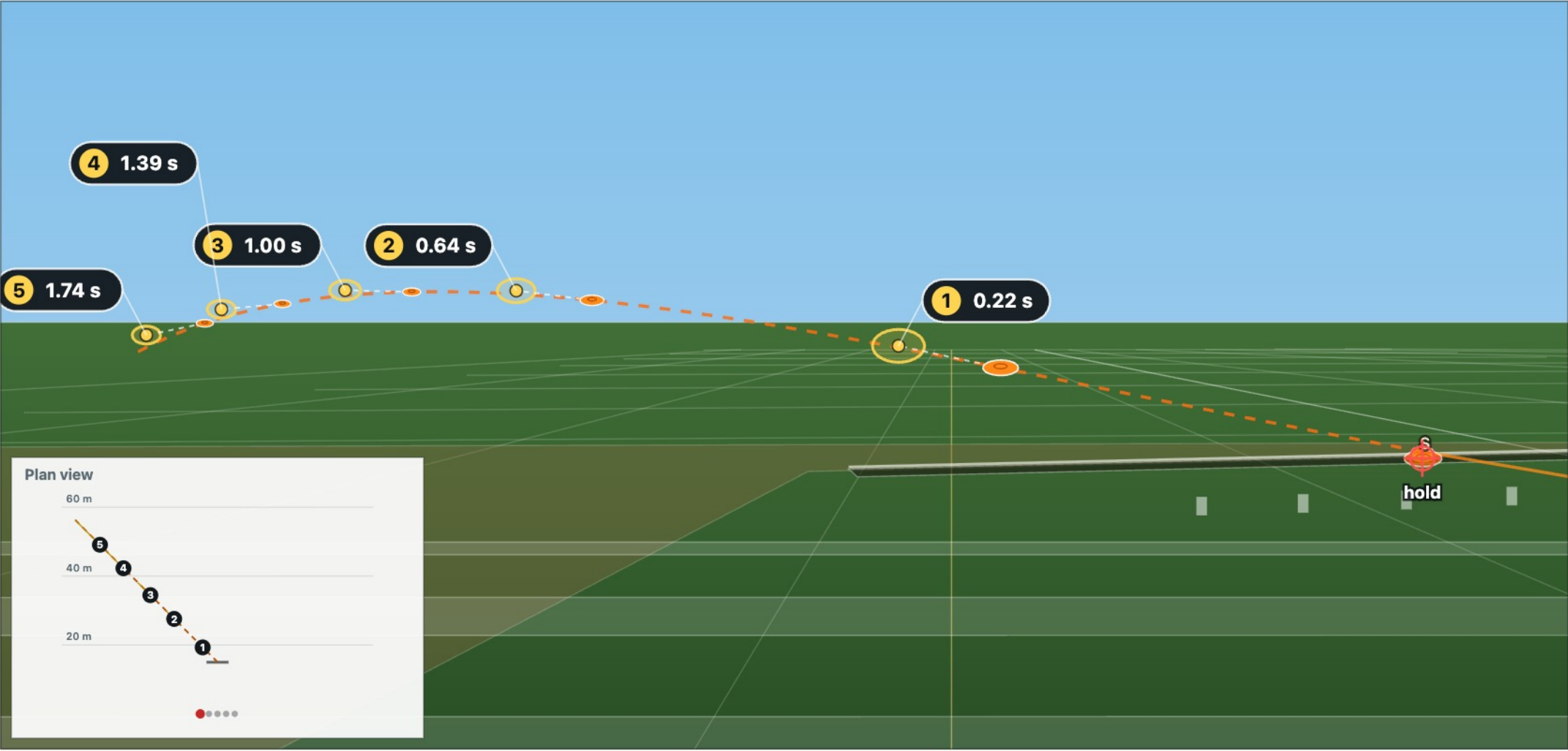
1 Release 0.22 s after release 102 cm ≈ 9.3 clays 93 cm left · 43 cm up -2.80° lateral · +1.30° vertical clay at 19 m · height 1.5 m · real separation 1.6 m first glimpse · within reach	2 Early 0.78 s after release 105 cm ≈ 9.5 clays 103 cm left · 21 cm up -2.05° lateral · +0.41° vertical clay at 29 m · height 4.1 m · real separation 2.5 m quick shot · within reach	3 Good window 1.25 s after release 113 cm ≈ 10.2 clays 112 cm left · 12 cm down -1.71° lateral · -0.18° vertical clay at 38 m · height 5.3 m · real separation 3.5 m the comfortable shot · within reach	4 At the limit 1.78 s after release 141 cm ≈ 12.8 clays 123 cm left · 68 cm down -1.48° lateral · -0.81° vertical clay at 48 m · height 5.6 m · real separation 5.0 m last useful shot · at the limit	5 You are late 2.45 s after release 227 cm ≈ 20.6 clays 141 cm left · 178 cm down -1.33° lateral · -1.69° vertical clay at 61 m · height 4.2 m · real separation 7.4 m no longer reaching well · beyond useful range
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Station 1 · Machine 5

angle -45° · height 1.5 m at 10 m · fall 60 m · 101 km/h · elevation 9.1°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned -26° horizontally and +4° vertically



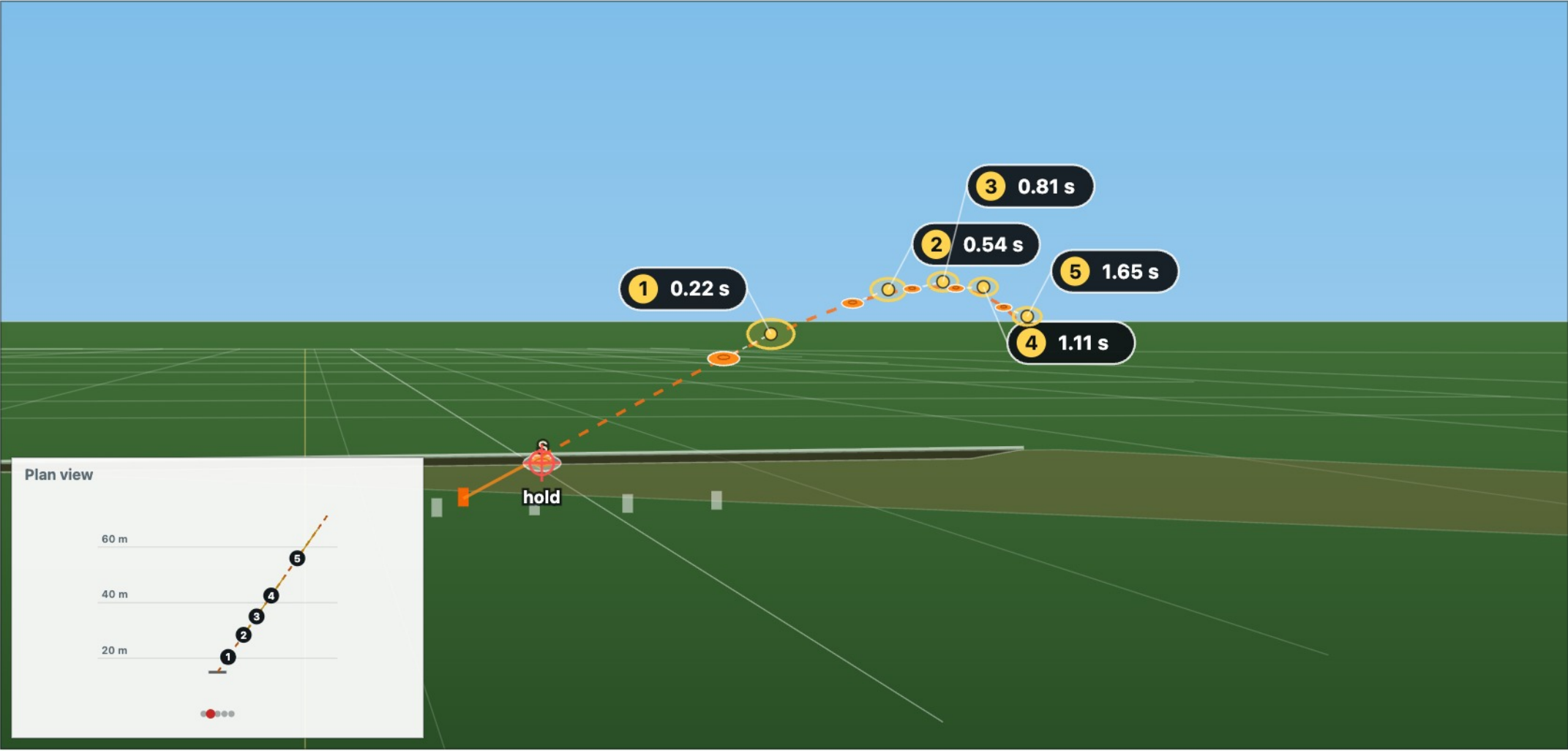
1 Release 0.22 s after release 160 cm ≈ 14.6 clays 157 cm left · 31 cm up -4.65° lateral · +0.93° vertical clay at 19 m · height 1.0 m · real separation 2.2 m first glimpse · within reach	2 Early 0.64 s after release 176 cm ≈ 16.0 clays 175 cm left · 15 cm up -3.50° lateral · +0.29° vertical clay at 29 m · height 2.2 m · real separation 3.5 m quick shot · within reach	3 Good window 1.00 s after release 193 cm ≈ 17.5 clays 193 cm left · 11 cm down -2.95° lateral · -0.17° vertical clay at 37 m · height 2.5 m · real separation 5.1 m the comfortable shot · within reach	4 At the limit 1.39 s after release 224 cm ≈ 20.4 clays 216 cm left · 58 cm down -2.59° lateral · -0.70° vertical clay at 48 m · height 2.4 m · real separation 7.3 m last useful shot · at the limit	5 You are late 1.74 s after release 270 cm ≈ 24.6 clays 241 cm left · 121 cm down -2.42° lateral · -1.22° vertical clay at 57 m · height 1.7 m · real separation 9.8 m no longer reaching well · beyond useful range
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Station 2 · Machine 1

angle +35° · height 1.5 m at 10 m · fall 70 m · 110 km/h · elevation 8.9°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned +10° horizontally and +4° vertically



Plan view



1 Release
0.22 s after release

94 cm

≈ 8.6 clays

85 cm right · 41 cm up

+2.28° lateral · +1.09° vertical

clay at 21 m · height 1.1 m · real separation 2.8 m

first glimpse · within reach

2 Early
0.54 s after release

97 cm

≈ 8.9 clays

94 cm right · 26 cm up

+1.74° lateral · +0.49° vertical

clay at 31 m · height 2.1 m · real separation 4.4 m

quick shot · within reach

3 Good window
0.81 s after release

102 cm

≈ 9.3 clays

102 cm right · 5 cm up

+1.51° lateral · +0.08° vertical

clay at 39 m · height 2.6 m · real separation 6.0 m

the comfortable shot · at the limit

4 At the limit
1.11 s after release

117 cm

≈ 10.7 clays

113 cm right · 31 cm down

+1.36° lateral · -0.37° vertical

clay at 48 m · height 2.9 m · real separation 8.3 m

last useful shot · at the limit

5 You are late
1.65 s after release

188 cm

≈ 17.1 clays

130 cm right · 136 cm down

+1.16° lateral · -1.22° vertical

clay at 64 m · height 2.4 m · real separation 12.8 m

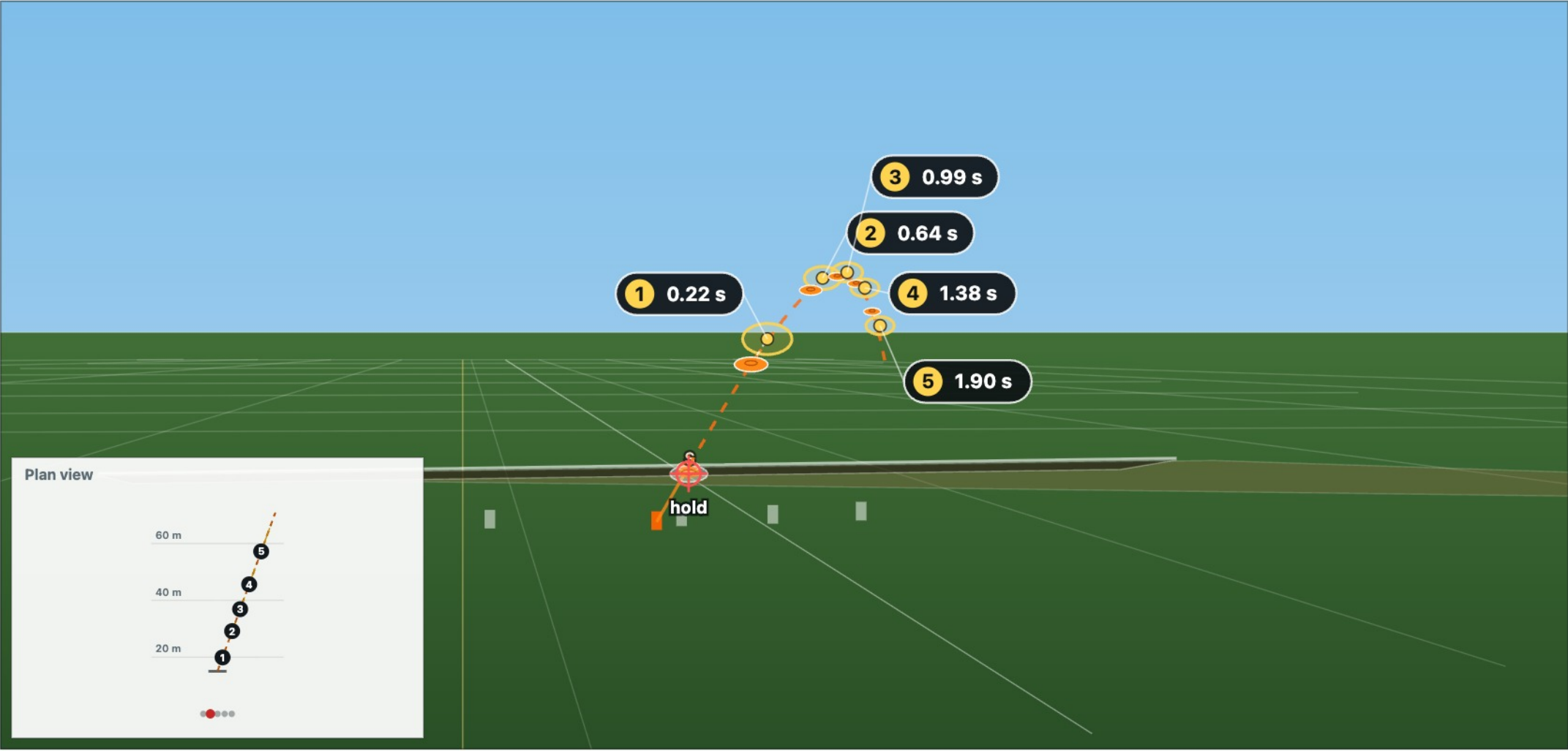
no longer reaching well · beyond useful range

Station 2 · Machine 2

angle +20° · height 2.0 m at 10 m · fall 60 m · 87 km/h · elevation 12.5°

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cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned +4° horizontally and +4° vertically



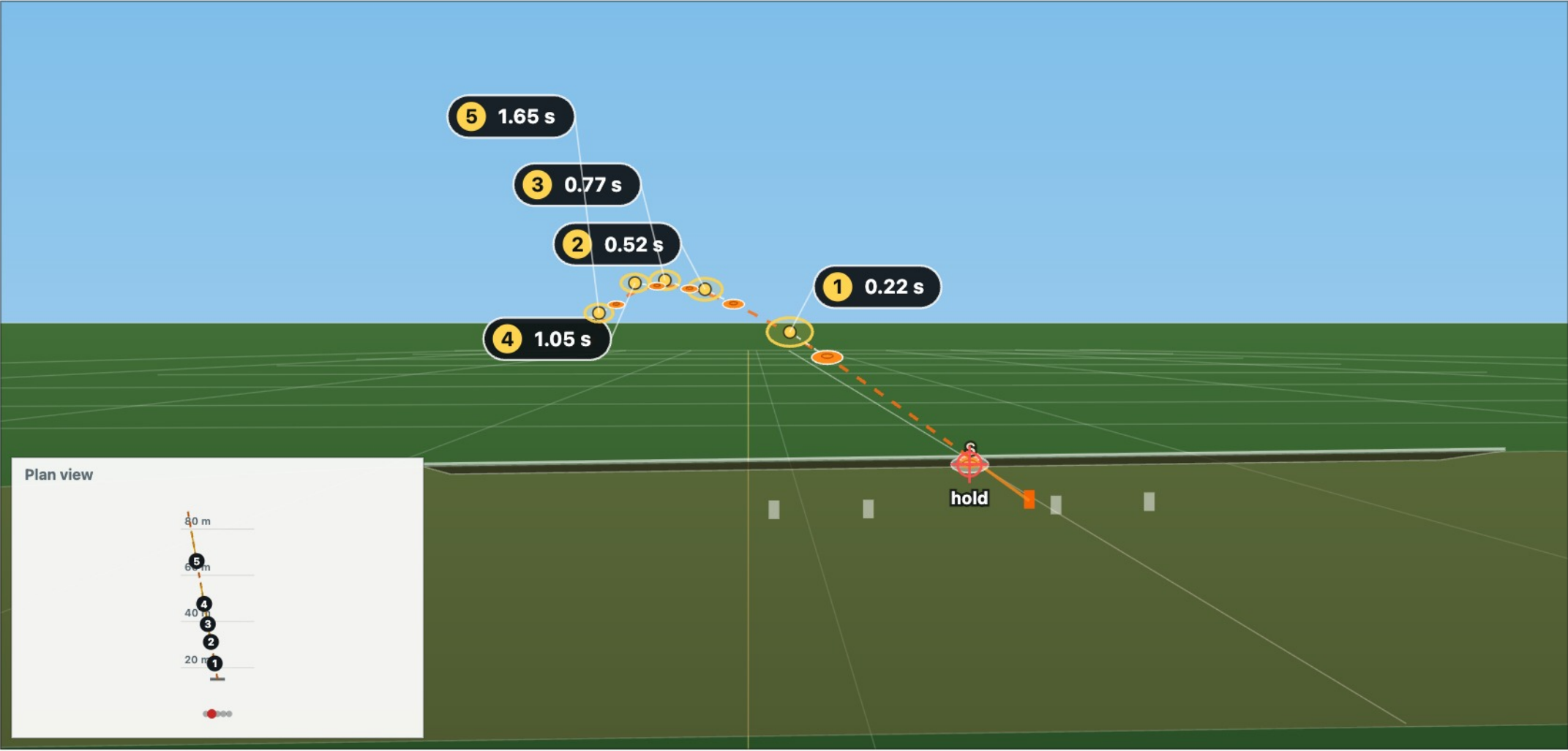
1 Release 0.22 s after release 48 cm ≈ 4.4 clays 27 cm right · 39 cm up +0.76° lateral · +1.11° vertical clay at 20 m · height 1.2 m · real separation 2.0 m first glimpse · within reach	2 Early 0.64 s after release 37 cm ≈ 3.3 clays 30 cm right · 21 cm up +0.56° lateral · +0.41° vertical clay at 30 m · height 2.6 m · real separation 3.2 m quick shot · within reach	3 Good window 0.99 s after release 32 cm ≈ 2.9 clays 32 cm right · 4 cm down +0.48° lateral · -0.06° vertical clay at 38 m · height 3.3 m · real separation 4.4 m the comfortable shot · at the limit	4 At the limit 1.38 s after release 59 cm ≈ 5.4 clays 35 cm right · 48 cm down +0.43° lateral · -0.57° vertical clay at 48 m · height 3.4 m · real separation 6.1 m last useful shot · at the limit	5 You are late 1.90 s after release 144 cm ≈ 13.1 clays 41 cm right · 139 cm down +0.39° lateral · -1.32° vertical clay at 60 m · height 2.6 m · real separation 8.8 m no longer reaching well · beyond useful range
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Station 2 · Machine 3

angle -10° · height 1.5 m at 10 m · fall 75 m · 115 km/h · elevation 8.8°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned -8° horizontally and +4° vertically



Plan view



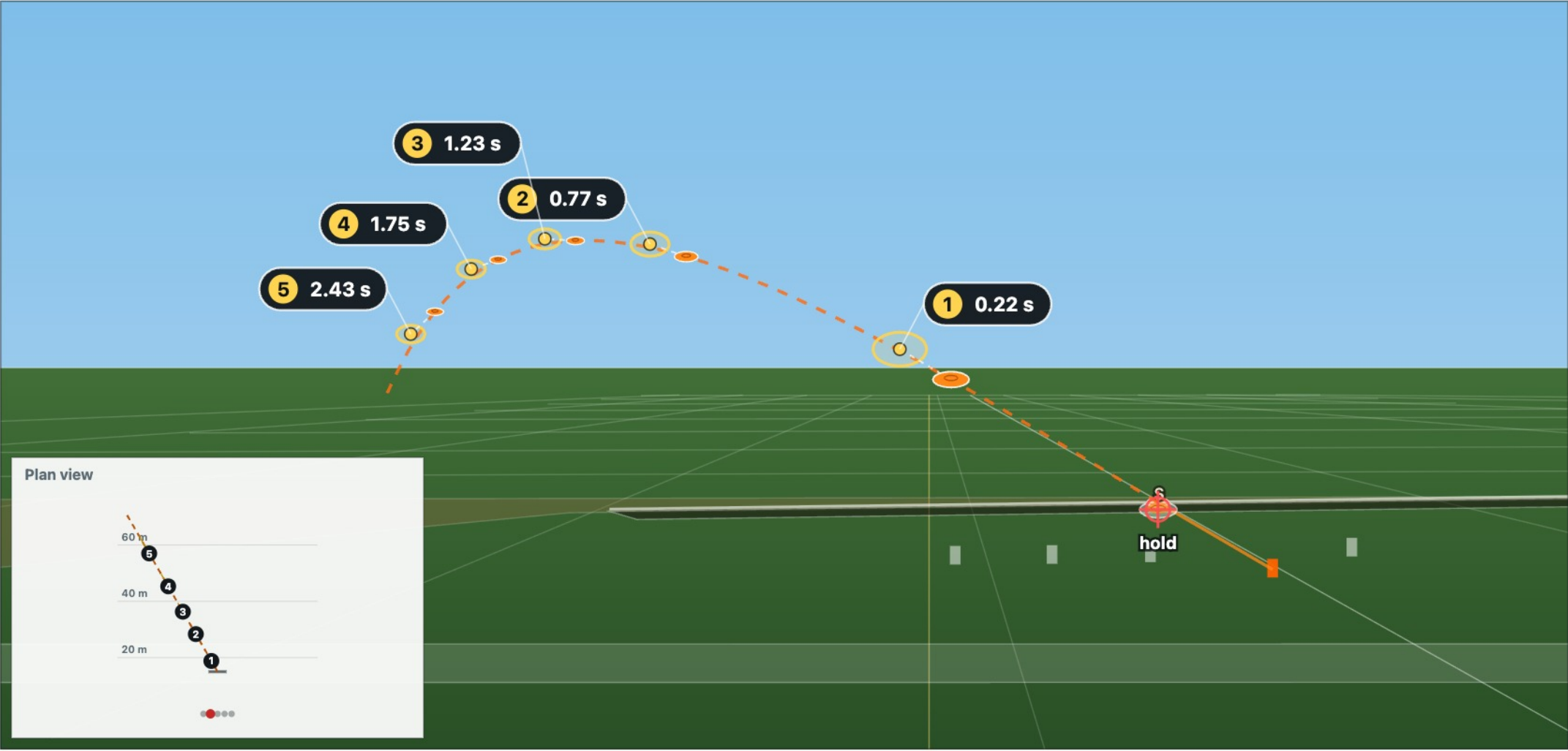
1 Release 0.22 s after release 82 cm ≈ 7.4 clays 69 cm left · 43 cm up -1.82° lateral · +1.14° vertical clay at 22 m · height 1.1 m · real separation 3.0 m first glimpse · within reach	2 Early 0.52 s after release 82 cm ≈ 7.5 clays 76 cm left · 30 cm up -1.41° lateral · +0.55° vertical clay at 31 m · height 2.1 m · real separation 4.7 m quick shot · within reach	3 Good window 0.77 s after release 84 cm ≈ 7.6 clays 83 cm left · 10 cm up -1.23° lateral · +0.15° vertical clay at 39 m · height 2.7 m · real separation 6.4 m the comfortable shot · at the limit	4 At the limit 1.05 s after release 95 cm ≈ 8.7 clays 92 cm left · 24 cm down -1.11° lateral · -0.29° vertical clay at 48 m · height 3.0 m · real separation 8.7 m last useful shot · at the limit	5 You are late 1.65 s after release 168 cm ≈ 15.3 clays 101 cm left · 134 cm down -0.87° lateral · -1.16° vertical clay at 67 m · height 2.6 m · real separation 13.3 m no longer reaching well · beyond useful range
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Station 2 · Machine 4

angle -30° · height 3.0 m at 10 m · fall 65 m · 76 km/h · elevation 18.6°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned -16° horizontally and +6° vertically



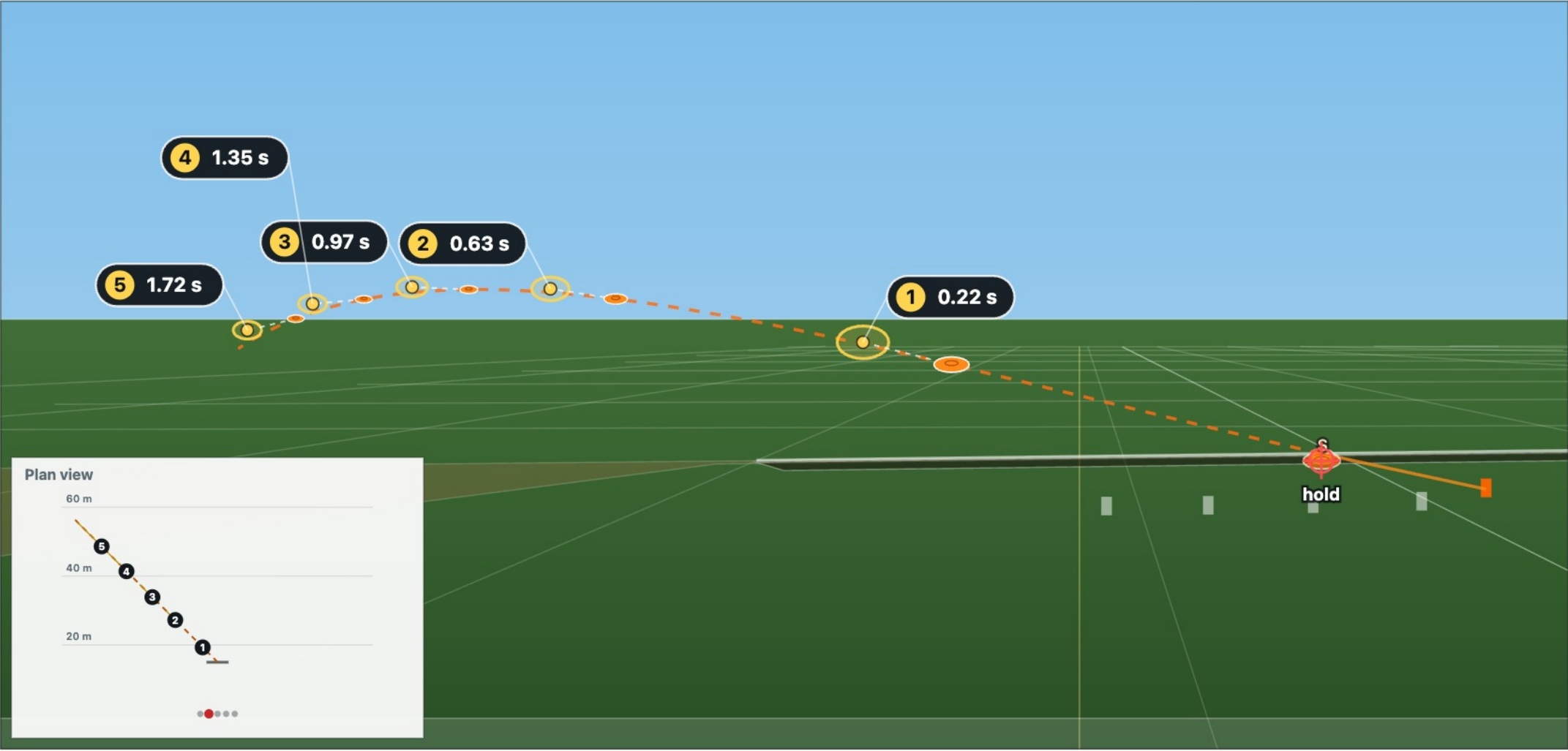
1 Release 0.22 s after release 88 cm ≈ 8.0 clays 76 cm left · 43 cm up -2.32° lateral · +1.31° vertical clay at 19 m · height 1.5 m · real separation 1.5 m first glimpse · within reach	2 Early 0.77 s after release 87 cm ≈ 7.9 clays 84 cm left · 21 cm up -1.67° lateral · +0.42° vertical clay at 29 m · height 4.1 m · real separation 2.5 m quick shot · within reach	3 Good window 1.23 s after release 92 cm ≈ 8.4 clays 92 cm left · 11 cm down -1.39° lateral · -0.16° vertical clay at 38 m · height 5.3 m · real separation 3.6 m the comfortable shot · within reach	4 At the limit 1.75 s after release 121 cm ≈ 11.0 clays 101 cm left · 66 cm down -1.21° lateral · -0.79° vertical clay at 48 m · height 5.6 m · real separation 5.0 m last useful shot · at the limit	5 You are late 2.43 s after release 214 cm ≈ 19.4 clays 116 cm left · 180 cm down -1.09° lateral · -1.68° vertical clay at 61 m · height 4.3 m · real separation 7.5 m no longer reaching well · beyond useful range
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Station 2 · Machine 5

angle -45° · height 1.5 m at 10 m · fall 60 m · 101 km/h · elevation 9.1°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned -22° horizontally and +4° vertically



Plan view



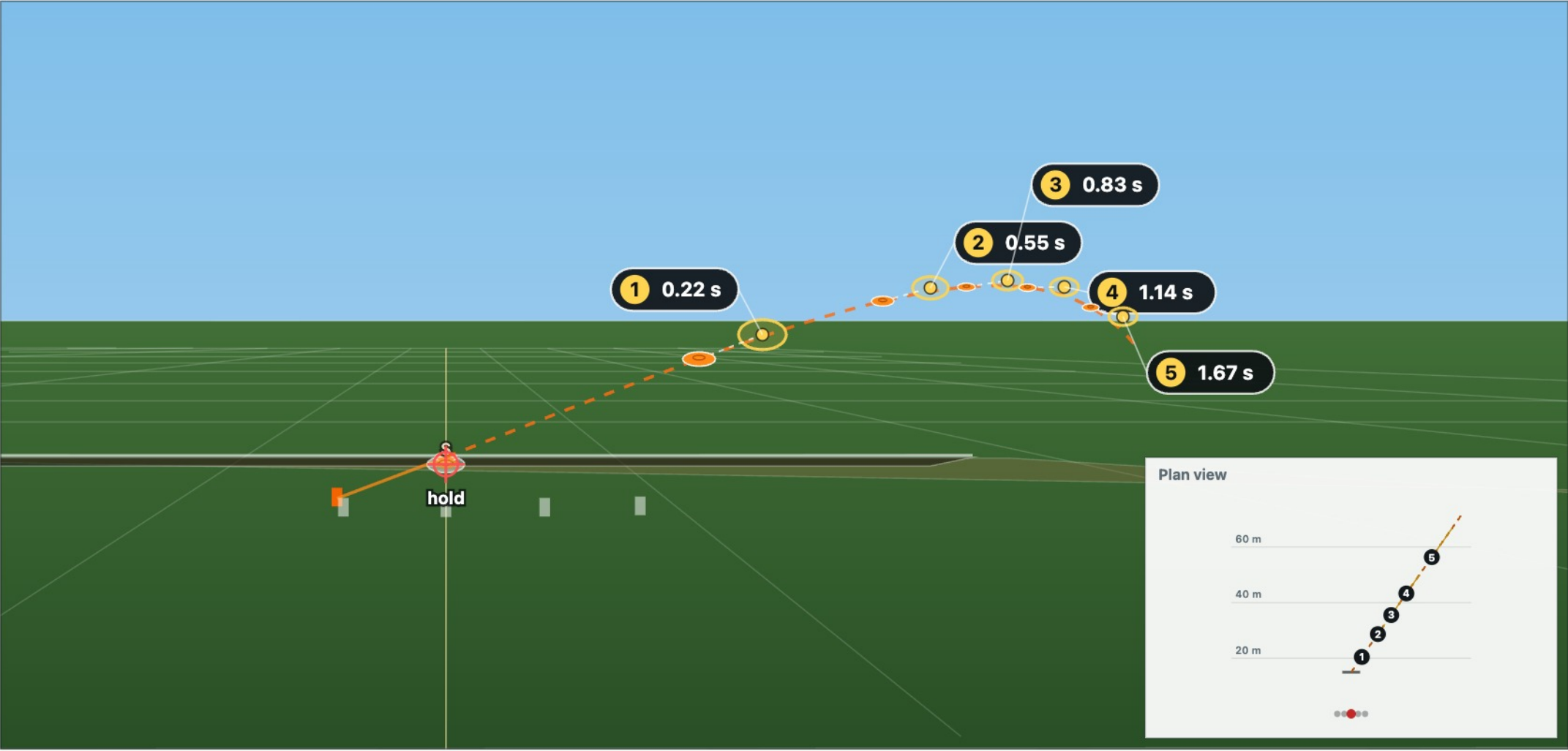
1 Release 0.22 s after release 143 cm ≈ 13.0 clays 139 cm left · 32 cm up -4.10° lateral · +0.95° vertical clay at 19 m · height 1.0 m · real separation 2.2 m first glimpse · within reach	2 Early 0.63 s after release 155 cm ≈ 14.1 clays 155 cm left · 16 cm up -3.05° lateral · +0.32° vertical clay at 29 m · height 2.1 m · real separation 3.6 m quick shot · within reach	3 Good window 0.97 s after release 170 cm ≈ 15.4 clays 170 cm left · 9 cm down -2.58° lateral · -0.14° vertical clay at 38 m · height 2.5 m · real separation 5.2 m the comfortable shot · within reach	4 At the limit 1.35 s after release 197 cm ≈ 17.9 clays 190 cm left · 54 cm down -2.27° lateral · -0.65° vertical clay at 48 m · height 2.4 m · real separation 7.3 m last useful shot · at the limit	5 You are late 1.72 s after release 246 cm ≈ 22.3 clays 213 cm left · 122 cm down -2.12° lateral · -1.21° vertical clay at 58 m · height 1.7 m · real separation 10.0 m no longer reaching well · beyond useful range
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Station 3 · Machine 1

angle +35° · height 1.5 m at 10 m · fall 70 m · 110 km/h · elevation 8.9°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned +14° horizontally and +4° vertically



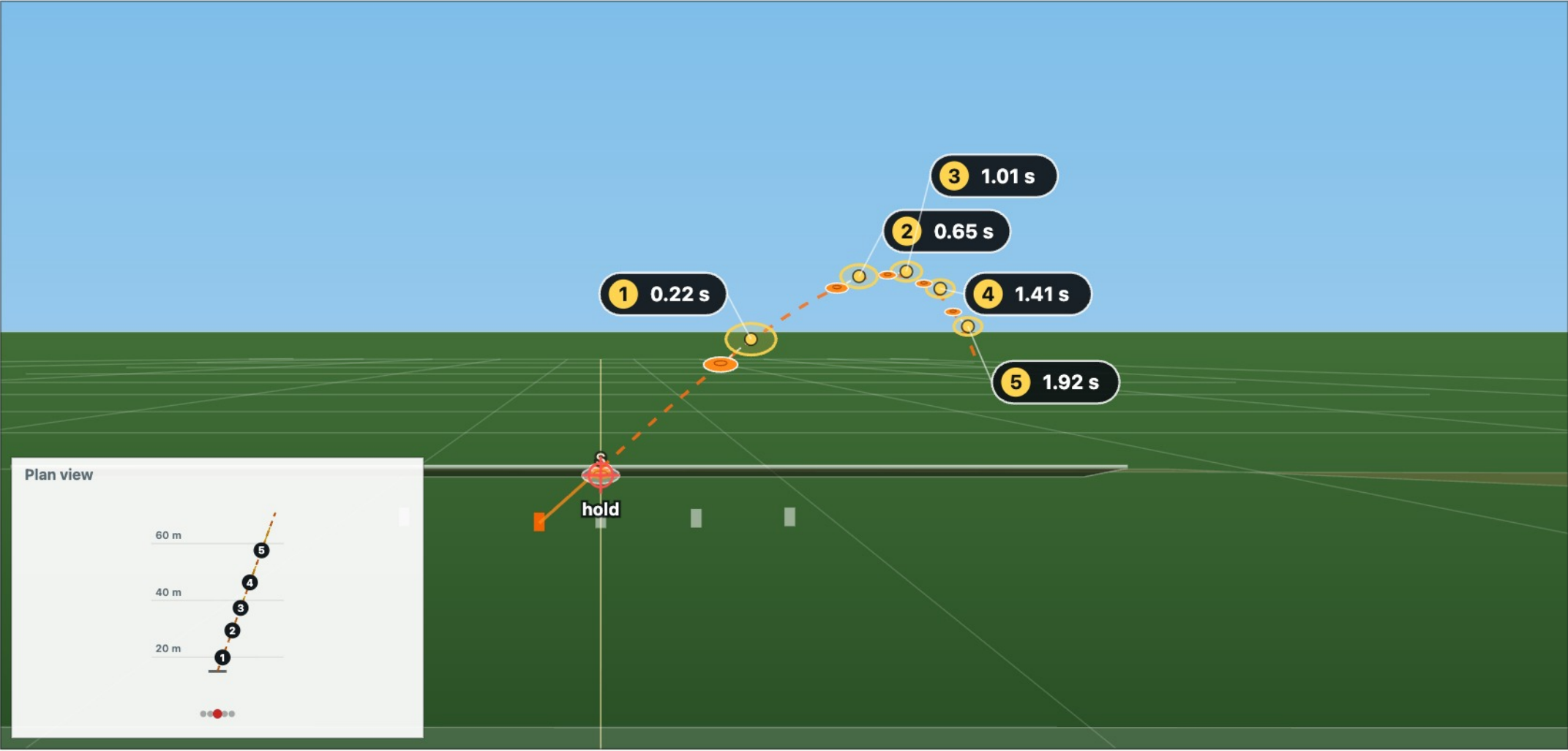
1 Release 0.22 s after release 117 cm ≈ 10.7 clays 110 cm right · 39 cm up +3.04° lateral · +1.08° vertical clay at 21 m · height 1.1 m · real separation 2.7 m first glimpse · within reach	2 Early 0.55 s after release 125 cm ≈ 11.3 clays 122 cm right · 25 cm up +2.31° lateral · +0.47° vertical clay at 30 m · height 2.1 m · real separation 4.3 m quick shot · within reach	3 Good window 0.83 s after release 133 cm ≈ 12.1 clays 133 cm right · 4 cm up +1.99° lateral · +0.05° vertical clay at 38 m · height 2.7 m · real separation 5.9 m the comfortable shot · at the limit	4 At the limit 1.14 s after release 152 cm ≈ 13.8 clays 148 cm right · 34 cm down +1.78° lateral · -0.41° vertical clay at 48 m · height 2.9 m · real separation 8.2 m last useful shot · at the limit	5 You are late 1.67 s after release 221 cm ≈ 20.1 clays 173 cm right · 138 cm down +1.56° lateral · -1.25° vertical clay at 63 m · height 2.3 m · real separation 12.8 m no longer reaching well · beyond useful range
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Station 3 · Machine 2

angle +20° · height 2.0 m at 10 m · fall 60 m · 87 km/h · elevation 12.5°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned +8° horizontally and +4° vertically



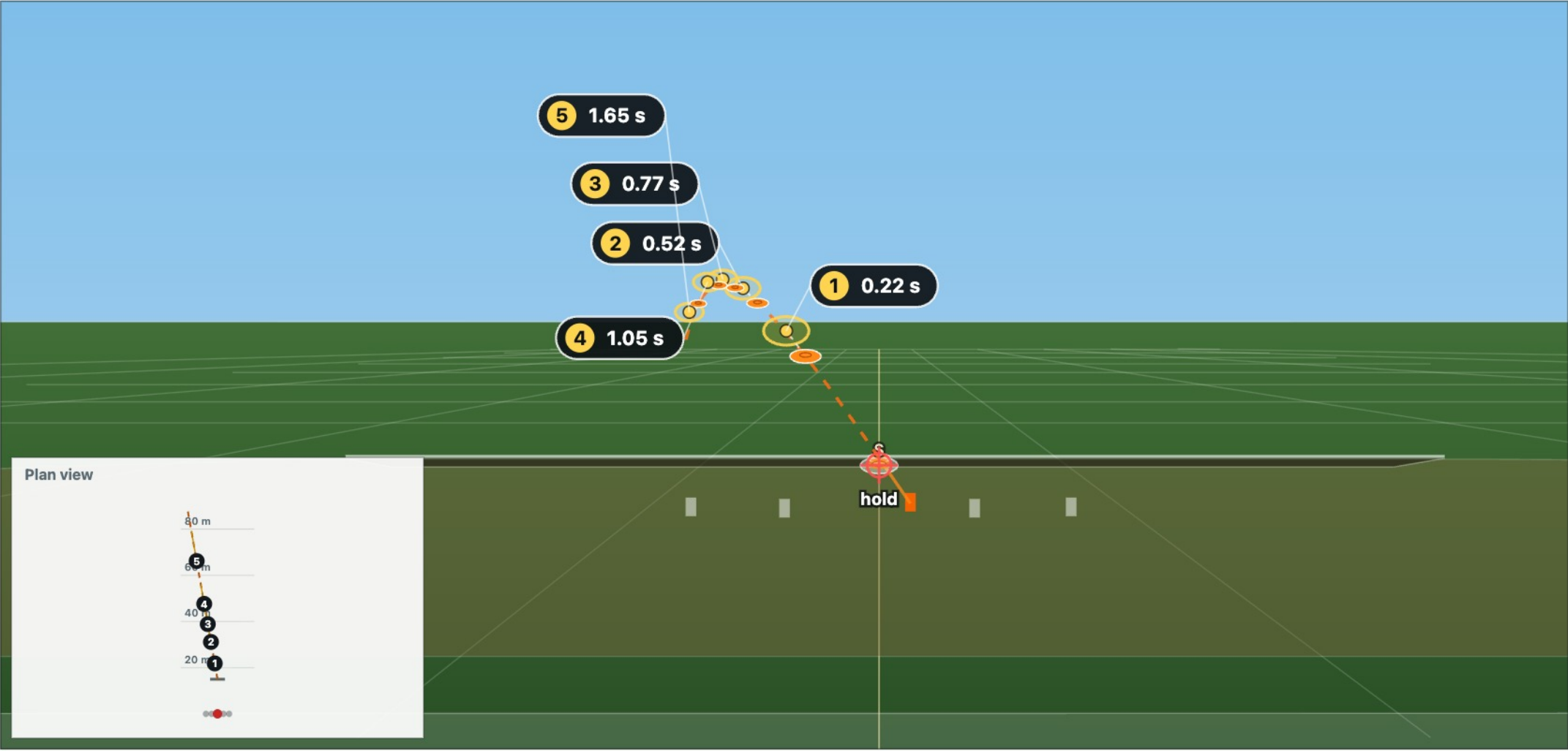
1 Release 0.22 s after release 63 cm ≈ 5.7 clays 50 cm right · 39 cm up +1.43° lateral · +1.10° vertical clay at 20 m · height 1.2 m · real separation 2.0 m first glimpse · within reach	2 Early 0.65 s after release 58 cm ≈ 5.3 clays 55 cm right · 20 cm up +1.05° lateral · +0.39° vertical clay at 30 m · height 2.7 m · real separation 3.2 m quick shot · within reach	3 Good window 1.01 s after release 60 cm ≈ 5.4 clays 59 cm right · 6 cm down +0.89° lateral · -0.09° vertical clay at 38 m · height 3.3 m · real separation 4.4 m the comfortable shot · at the limit	4 At the limit 1.41 s after release 83 cm ≈ 7.5 clays 65 cm right · 51 cm down +0.78° lateral · -0.61° vertical clay at 48 m · height 3.4 m · real separation 6.1 m last useful shot · at the limit	5 You are late 1.92 s after release 158 cm ≈ 14.3 clays 74 cm right · 139 cm down +0.72° lateral · -1.33° vertical clay at 60 m · height 2.6 m · real separation 8.7 m no longer reaching well · beyond useful range
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Station 3 · Machine 3

angle -10° · height 1.5 m at 10 m · fall 75 m · 115 km/h · elevation 8.8°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned -4° horizontally and +4° vertically



Plan view



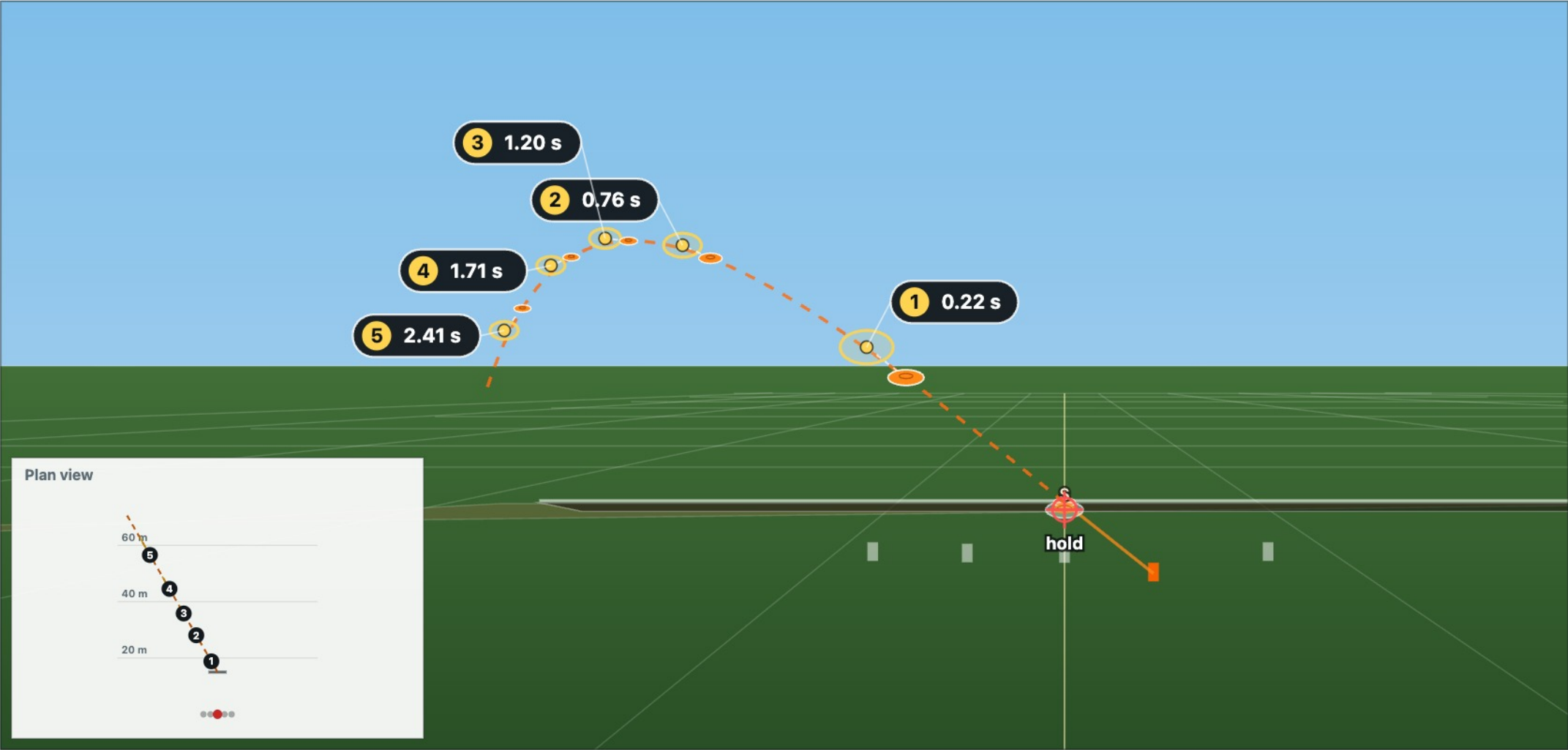
1 Release 0.22 s after release 57 cm ≈ 5.1 clays 36 cm left · 44 cm up -0.94° lateral · +1.14° vertical clay at 22 m · height 1.1 m · real separation 3.0 m first glimpse · within reach	2 Early 0.52 s after release 49 cm ≈ 4.5 clays 39 cm left · 30 cm up -0.72° lateral · +0.55° vertical clay at 31 m · height 2.1 m · real separation 4.7 m quick shot · within reach	3 Good window 0.77 s after release 44 cm ≈ 4.0 clays 43 cm left · 10 cm up -0.63° lateral · +0.15° vertical clay at 39 m · height 2.7 m · real separation 6.4 m the comfortable shot · at the limit	4 At the limit 1.05 s after release 53 cm ≈ 4.9 clays 48 cm left · 24 cm down -0.57° lateral · -0.29° vertical clay at 48 m · height 3.0 m · real separation 8.8 m last useful shot · at the limit	5 You are late 1.65 s after release 144 cm ≈ 13.1 clays 52 cm left · 134 cm down -0.44° lateral · -1.15° vertical clay at 67 m · height 2.6 m · real separation 13.3 m no longer reaching well · beyond useful range
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Station 3 · Machine 4

angle -30° · height 3.0 m at 10 m · fall 65 m · 76 km/h · elevation 18.6°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned -12° horizontally and +6° vertically



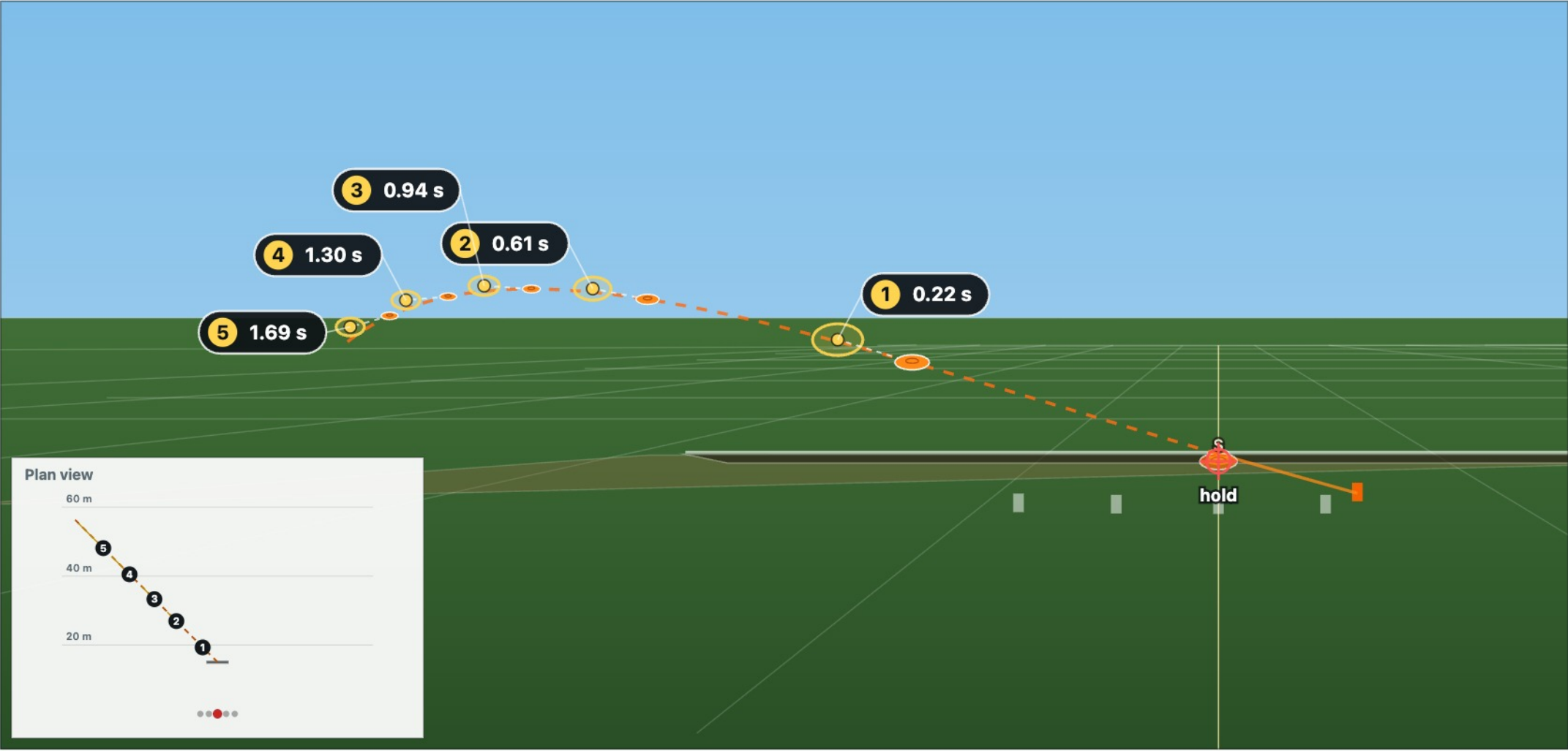
1 Release 0.22 s after release 74 cm ≈ 6.7 clays 60 cm left · 44 cm up -1.80° lateral · +1.32° vertical clay at 19 m · height 1.5 m · real separation 1.6 m first glimpse · within reach	2 Early 0.76 s after release 69 cm ≈ 6.3 clays 66 cm left · 22 cm up -1.29° lateral · +0.44° vertical clay at 29 m · height 4.0 m · real separation 2.6 m quick shot · within reach	3 Good window 1.20 s after release 72 cm ≈ 6.5 clays 71 cm left · 8 cm down -1.08° lateral · -0.13° vertical clay at 38 m · height 5.2 m · real separation 3.6 m the comfortable shot · within reach	4 At the limit 1.71 s after release 100 cm ≈ 9.1 clays 79 cm left · 62 cm down -0.94° lateral · -0.74° vertical clay at 48 m · height 5.6 m · real separation 5.0 m last useful shot · at the limit	5 You are late 2.41 s after release 201 cm ≈ 18.3 clays 90 cm left · 179 cm down -0.84° lateral · -1.67° vertical clay at 62 m · height 4.4 m · real separation 7.6 m no longer reaching well · beyond useful range
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Station 3 • Machine 5

angle -45° • height 1.5 m at 10 m • fall 60 m • 101 km/h • elevation 9.1°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s • rib 0% (0.00°) • wind 0.0 m/s • delay 0.03 s • pattern 1 3/4 • hold neutral
framing 1.58x over the base field of 48° vertical • view turned -18° horizontally and +4° vertically



Plan view



1 Release
0.22 s after release

125 cm

≈ 11.4 clays

121 cm left • 34 cm up

-3.49° lateral • +0.98° vertical

clay at 20 m • height 1.0 m • real separation 2.3 m

first glimpse • within reach

2 Early
0.61 s after release

135 cm

≈ 12.3 clays

134 cm left • 17 cm up

-2.60° lateral • +0.34° vertical

clay at 29 m • height 2.1 m • real separation 3.7 m

quick shot • within reach

3 Good window
0.94 s after release

146 cm

≈ 13.3 clays

146 cm left • 7 cm down

-2.20° lateral • -0.11° vertical

clay at 38 m • height 2.5 m • real separation 5.2 m

the comfortable shot • at the limit

4 At the limit
1.30 s after release

171 cm

≈ 15.5 clays

163 cm left • 51 cm down

-1.95° lateral • -0.61° vertical

clay at 48 m • height 2.5 m • real separation 7.4 m

last useful shot • at the limit

5 You are late
1.69 s after release

222 cm

≈ 20.2 clays

185 cm left • 123 cm down

-1.81° lateral • -1.20° vertical

clay at 58 m • height 1.8 m • real separation 10.3 m

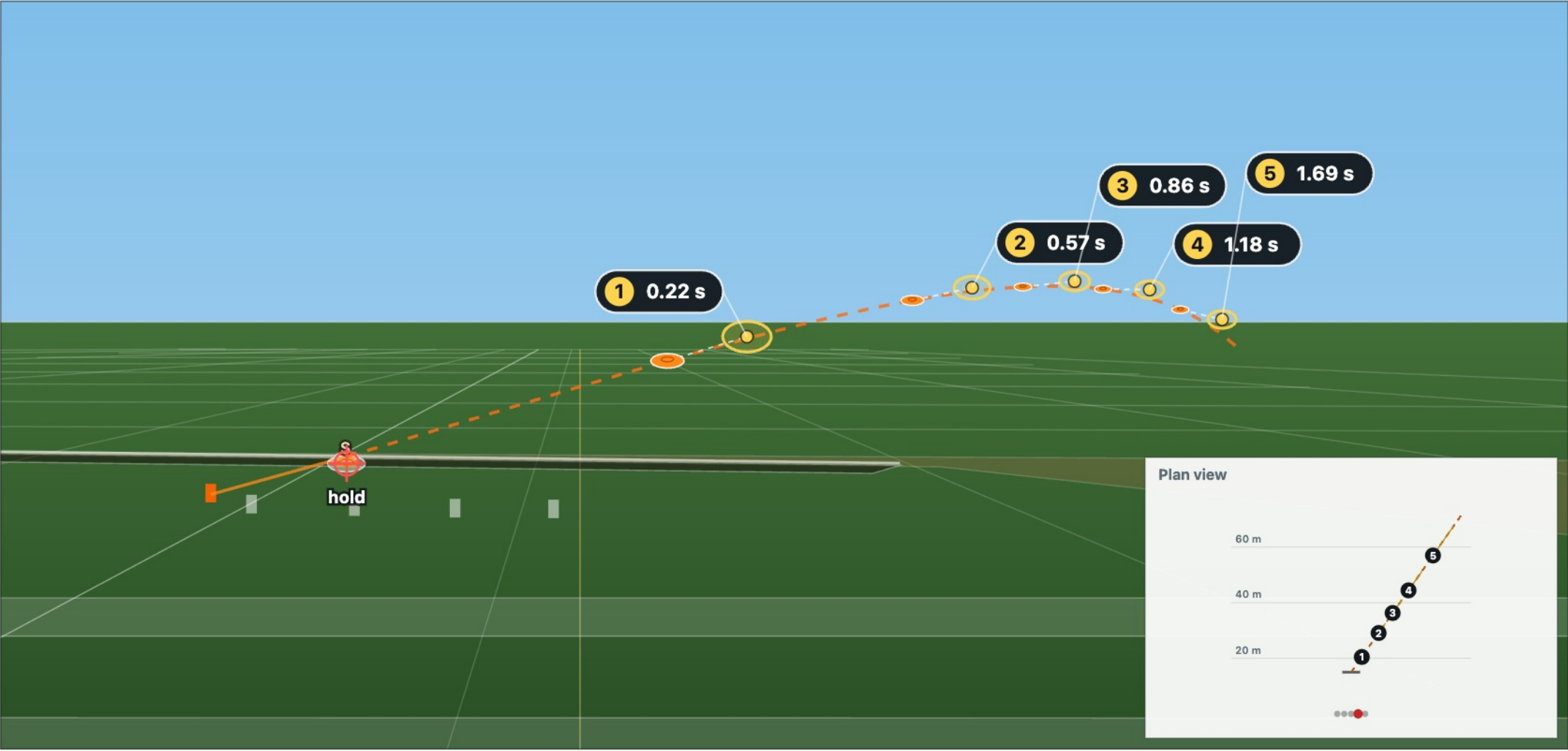
no longer reaching well • beyond useful range

Station 4 · Machine 1

angle +35° · height 1.5 m at 10 m · fall 70 m · 110 km/h · elevation 8.9°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned +18° horizontally and +4° vertically



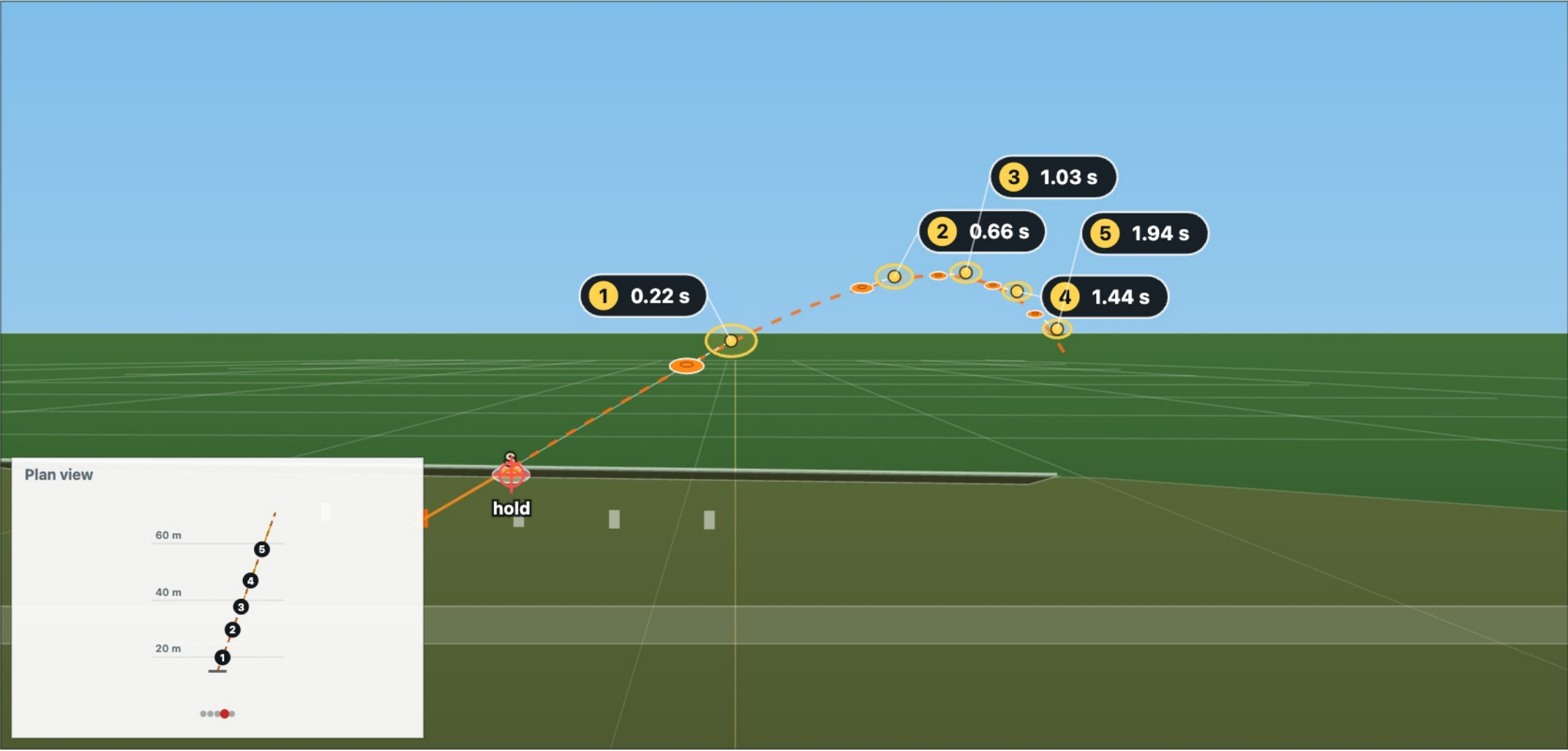
1 Release 0.22 s after release 141 cm ≈ 12.8 clays 135 cm right · 38 cm up +3.77° lateral · +1.06° vertical clay at 21 m · height 1.1 m · real separation 2.6 m first glimpse · within reach	2 Early 0.57 s after release 152 cm ≈ 13.8 clays 150 cm right · 23 cm up +2.87° lateral · +0.44° vertical clay at 30 m · height 2.2 m · real separation 4.2 m quick shot · within reach	3 Good window 0.86 s after release 165 cm ≈ 15.0 clays 165 cm right · 1 cm up +2.46° lateral · +0.01° vertical clay at 38 m · height 2.7 m · real separation 5.9 m the comfortable shot · at the limit	4 At the limit 1.18 s after release 188 cm ≈ 17.1 clays 184 cm right · 39 cm down +2.19° lateral · -0.47° vertical clay at 48 m · height 2.9 m · real separation 8.3 m last useful shot · at the limit	5 You are late 1.69 s after release 258 cm ≈ 23.4 clays 215 cm right · 142 cm down +1.96° lateral · -1.29° vertical clay at 63 m · height 2.3 m · real separation 12.8 m no longer reaching well · beyond useful range
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Station 4 · Machine 2

angle +20° · height 2.0 m at 10 m · fall 60 m · 87 km/h · elevation 12.5°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned +12° horizontally and +4° vertically



Plan view



1 Release 0.22 s after release 82 cm ≈ 7.4 clays 72 cm right · 38 cm up +2.07° lateral · +1.09° vertical clay at 20 m · height 1.2 m · real separation 1.9 m first glimpse · within reach	2 Early 0.66 s after release 82 cm ≈ 7.4 clays 79 cm right · 19 cm up +1.53° lateral · +0.37° vertical clay at 30 m · height 2.7 m · real separation 3.2 m quick shot · within reach	3 Good window 1.03 s after release 87 cm ≈ 7.9 clays 86 cm right · 7 cm down +1.29° lateral · -0.11° vertical clay at 38 m · height 3.3 m · real separation 4.4 m the comfortable shot · at the limit	4 At the limit 1.44 s after release 109 cm ≈ 9.9 clays 95 cm right · 54 cm down +1.14° lateral · -0.64° vertical clay at 48 m · height 3.4 m · real separation 6.1 m last useful shot · at the limit	5 You are late 1.94 s after release 177 cm ≈ 16.1 clays 108 cm right · 140 cm down +1.04° lateral · -1.35° vertical clay at 59 m · height 2.5 m · real separation 8.7 m no longer reaching well · beyond useful range
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Station 4 · Machine 3

angle -10° · height 1.5 m at 10 m · fall 75 m · 115 km/h · elevation 8.8°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned -0° horizontally and +4° vertically



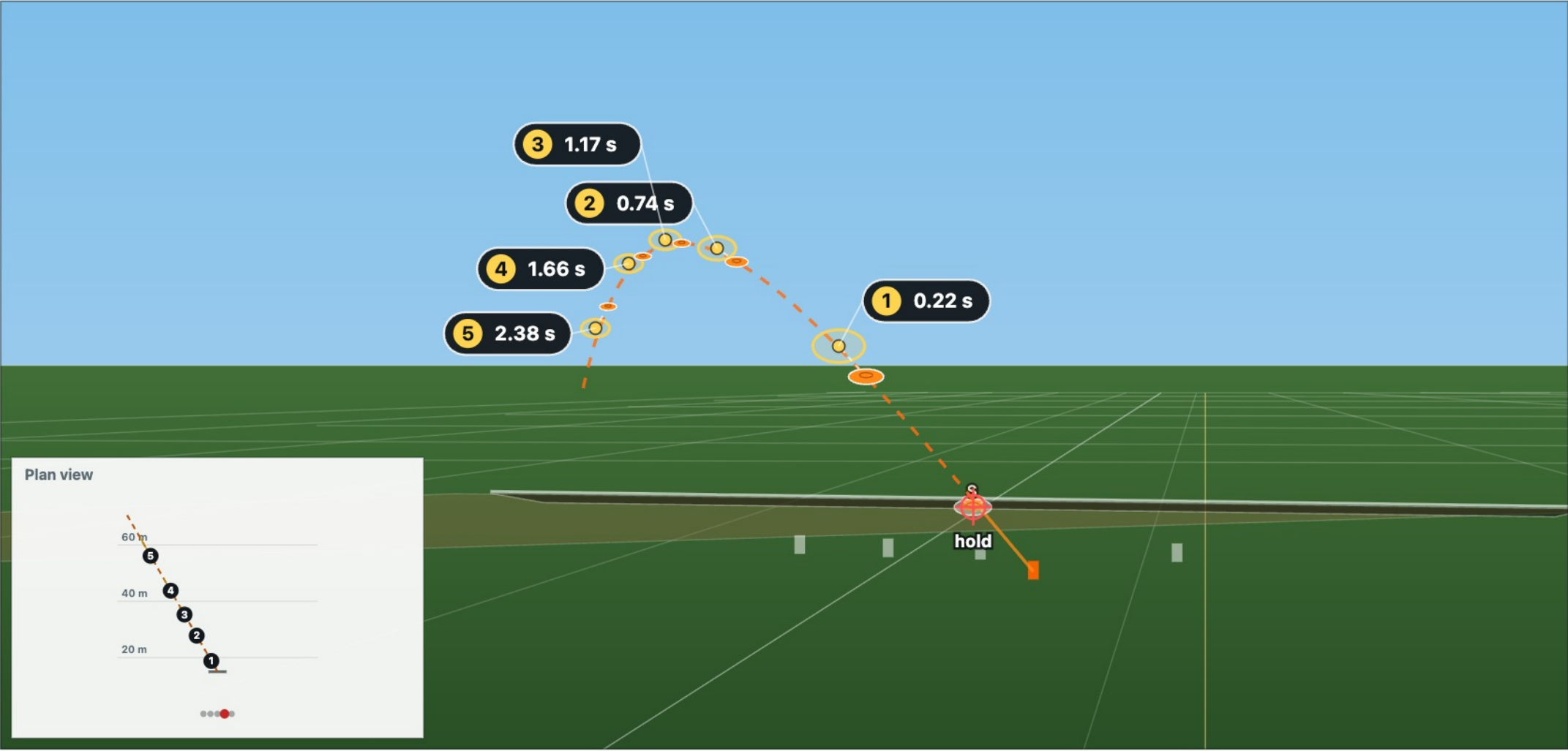
1 Release 0.22 s after release 44 cm ≈ 4.0 clays 2 cm left · 44 cm up -0.05° lateral · +1.15° vertical clay at 22 m · height 1.1 m · real separation 3.1 m first glimpse · within reach	2 Early 0.51 s after release 31 cm ≈ 2.8 clays 2 cm left · 31 cm up -0.04° lateral · +0.56° vertical clay at 31 m · height 2.1 m · real separation 4.8 m quick shot · within reach	3 Good window 0.76 s after release 11 cm ≈ 1.0 clays 2 cm left · 11 cm up -0.03° lateral · +0.16° vertical clay at 39 m · height 2.7 m · real separation 6.5 m the comfortable shot · at the limit	4 At the limit 1.04 s after release 23 cm ≈ 2.1 clays 3 cm left · 23 cm down -0.03° lateral · -0.27° vertical clay at 48 m · height 3.0 m · real separation 8.7 m last useful shot · at the limit	5 You are late 1.64 s after release 133 cm ≈ 12.1 clays 3 cm left · 133 cm down -0.02° lateral · -1.14° vertical clay at 67 m · height 2.6 m · real separation 13.3 m no longer reaching well · beyond useful range
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Station 4 · Machine 4

angle -30° · height 3.0 m at 10 m · fall 65 m · 76 km/h · elevation 18.6°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned -8° horizontally and +6° vertically



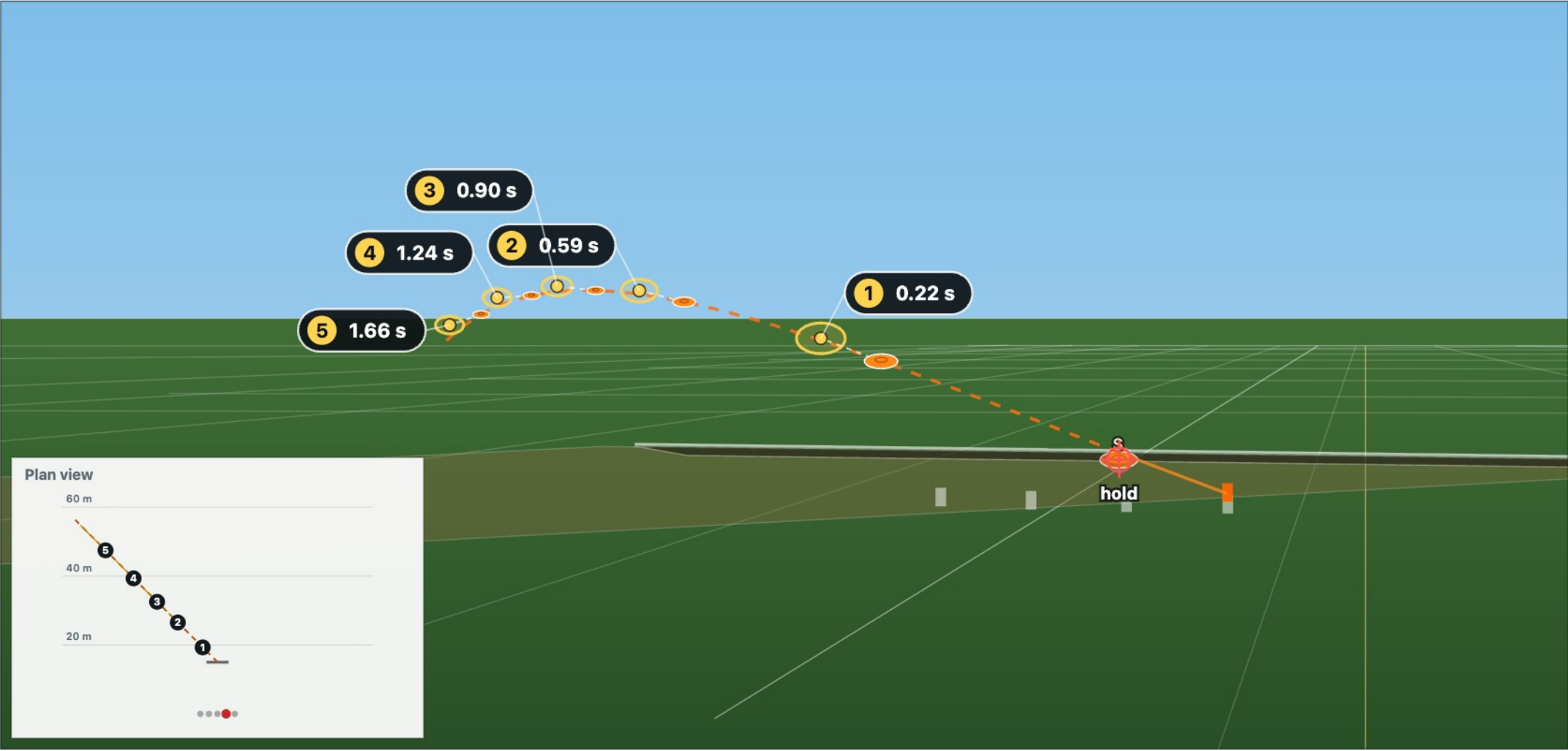
1 Release 0.22 s after release 62 cm ≈ 5.6 clays 43 cm left · 45 cm up -1.26° lateral · +1.33° vertical clay at 19 m · height 1.5 m · real separation 1.6 m first glimpse · within reach	2 Early 0.74 s after release 53 cm ≈ 4.8 clays 47 cm left · 24 cm up -0.91° lateral · +0.47° vertical clay at 30 m · height 4.0 m · real separation 2.6 m quick shot · within reach	3 Good window 1.17 s after release 51 cm ≈ 4.7 clays 51 cm left · 6 cm down -0.76° lateral · -0.09° vertical clay at 38 m · height 5.2 m · real separation 3.6 m the comfortable shot · at the limit	4 At the limit 1.66 s after release 80 cm ≈ 7.3 clays 56 cm left · 58 cm down -0.67° lateral · -0.69° vertical clay at 48 m · height 5.6 m · real separation 5.0 m last useful shot · at the limit	5 You are late 2.38 s after release 191 cm ≈ 17.3 clays 65 cm left · 179 cm down -0.60° lateral · -1.66° vertical clay at 62 m · height 4.4 m · real separation 7.7 m no longer reaching well · beyond useful range
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Station 4 • Machine 5

angle -45° • height 1.5 m at 10 m • fall 60 m • 101 km/h • elevation 9.1°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s • rib 0% (0.00°) • wind 0.0 m/s • delay 0.03 s • pattern 1 3/4 • hold neutral
framing 1.58x over the base field of 48° vertical • view turned -14° horizontally and +4° vertically



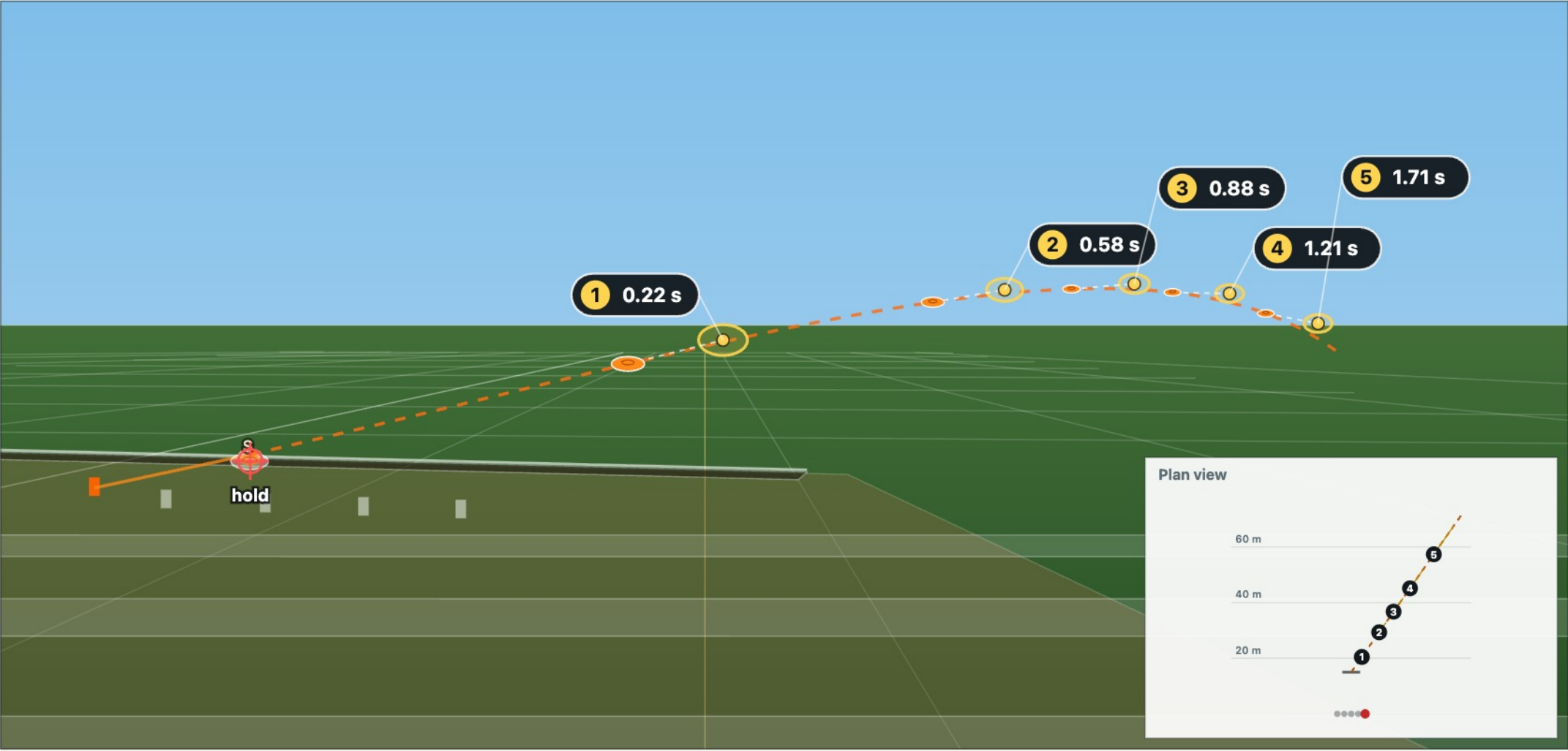
1 Release 0.22 s after release 108 cm ≈ 9.8 clays 102 cm left • 36 cm up -2.85° lateral • +1.00° vertical clay at 20 m • height 1.0 m • real separation 2.4 m first glimpse • within reach	2 Early 0.59 s after release 114 cm ≈ 10.4 clays 112 cm left • 20 cm up -2.14° lateral • +0.37° vertical clay at 30 m • height 2.1 m • real separation 3.8 m quick shot • within reach	3 Good window 0.90 s after release 122 cm ≈ 11.1 clays 122 cm left • 4 cm down -1.83° lateral • -0.06° vertical clay at 38 m • height 2.5 m • real separation 5.3 m the comfortable shot • at the limit	4 At the limit 1.24 s after release 143 cm ≈ 13.0 clays 136 cm left • 45 cm down -1.63° lateral • -0.54° vertical clay at 48 m • height 2.5 m • real separation 7.3 m last useful shot • at the limit	5 You are late 1.66 s after release 197 cm ≈ 17.9 clays 155 cm left • 121 cm down -1.51° lateral • -1.18° vertical clay at 59 m • height 1.9 m • real separation 10.4 m no longer reaching well • beyond useful range
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Station 5 · Machine 1

angle +35° · height 1.5 m at 10 m · fall 70 m · 110 km/h · elevation 8.9°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned +22° horizontally and +4° vertically



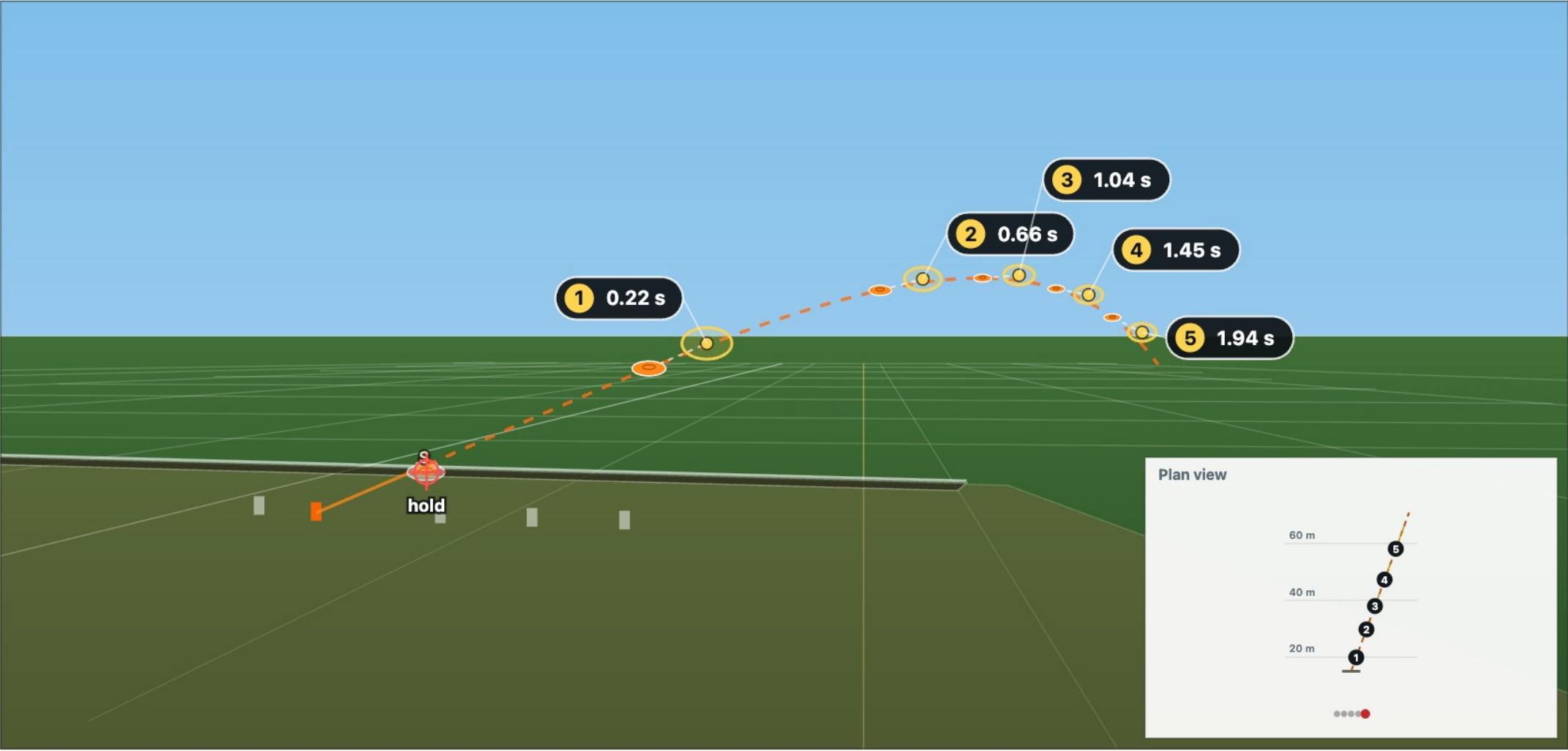
1 Release 0.22 s after release 164 cm ≈ 14.9 clays 160 cm right · 37 cm up +4.46° lateral · +1.03° vertical clay at 21 m · height 1.1 m · real separation 2.6 m first glimpse · within reach	2 Early 0.58 s after release 179 cm ≈ 16.3 clays 178 cm right · 22 cm up +3.42° lateral · +0.42° vertical clay at 30 m · height 2.2 m · real separation 4.2 m quick shot · within reach	3 Good window 0.88 s after release 195 cm ≈ 17.7 clays 195 cm right · 1 cm down +2.93° lateral · -0.01° vertical clay at 38 m · height 2.7 m · real separation 5.9 m the comfortable shot · at the limit	4 At the limit 1.21 s after release 223 cm ≈ 20.2 clays 219 cm right · 42 cm down +2.61° lateral · -0.50° vertical clay at 48 m · height 2.9 m · real separation 8.3 m last useful shot · at the limit	5 You are late 1.71 s after release 296 cm ≈ 26.9 clays 259 cm right · 144 cm down +2.37° lateral · -1.32° vertical clay at 62 m · height 2.2 m · real separation 12.8 m no longer reaching well · beyond useful range
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Station 5 · Machine 2

angle +20° · height 2.0 m at 10 m · fall 60 m · 87 km/h · elevation 12.5°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned +15° horizontally and +4° vertically



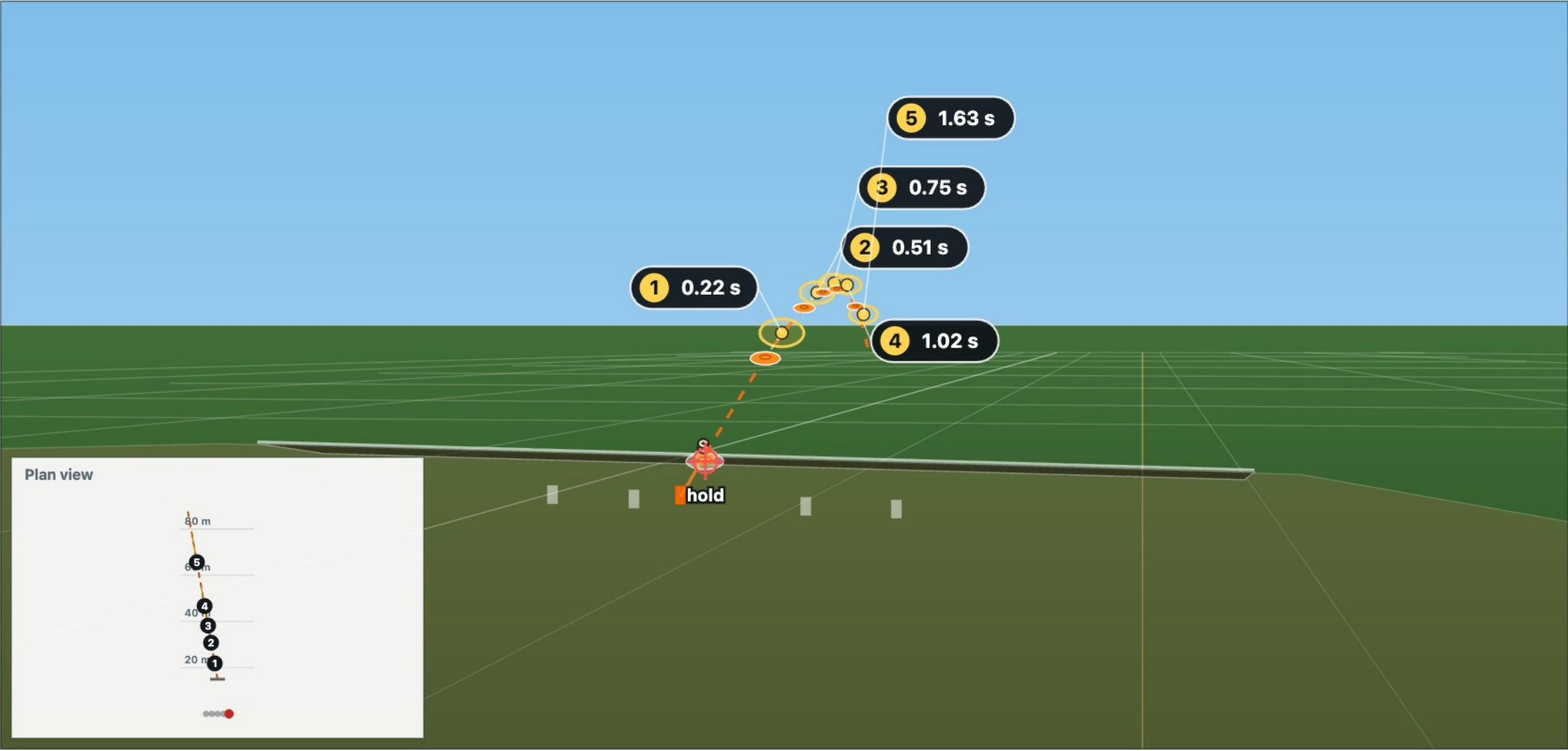
1 Release 0.22 s after release 102 cm ≈ 9.3 clays 94 cm right · 38 cm up +2.68° lateral · +1.08° vertical clay at 20 m · height 1.2 m · real separation 2.0 m first glimpse · within reach	2 Early 0.66 s after release 106 cm ≈ 9.6 clays 104 cm right · 19 cm up +2.00° lateral · +0.37° vertical clay at 30 m · height 2.7 m · real separation 3.2 m quick shot · within reach	3 Good window 1.04 s after release 113 cm ≈ 10.3 clays 113 cm right · 8 cm down +1.70° lateral · -0.12° vertical clay at 38 m · height 3.4 m · real separation 4.4 m the comfortable shot · at the limit	4 At the limit 1.45 s after release 136 cm ≈ 12.4 clays 125 cm right · 55 cm down +1.50° lateral · -0.65° vertical clay at 48 m · height 3.4 m · real separation 6.1 m last useful shot · at the limit	5 You are late 1.94 s after release 198 cm ≈ 18.0 clays 141 cm right · 139 cm down +1.37° lateral · -1.35° vertical clay at 59 m · height 2.5 m · real separation 8.6 m no longer reaching well · beyond useful range
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Station 5 · Machine 3

angle -10° · height 1.5 m at 10 m · fall 75 m · 115 km/h · elevation 8.8°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned +3° horizontally and +4° vertically



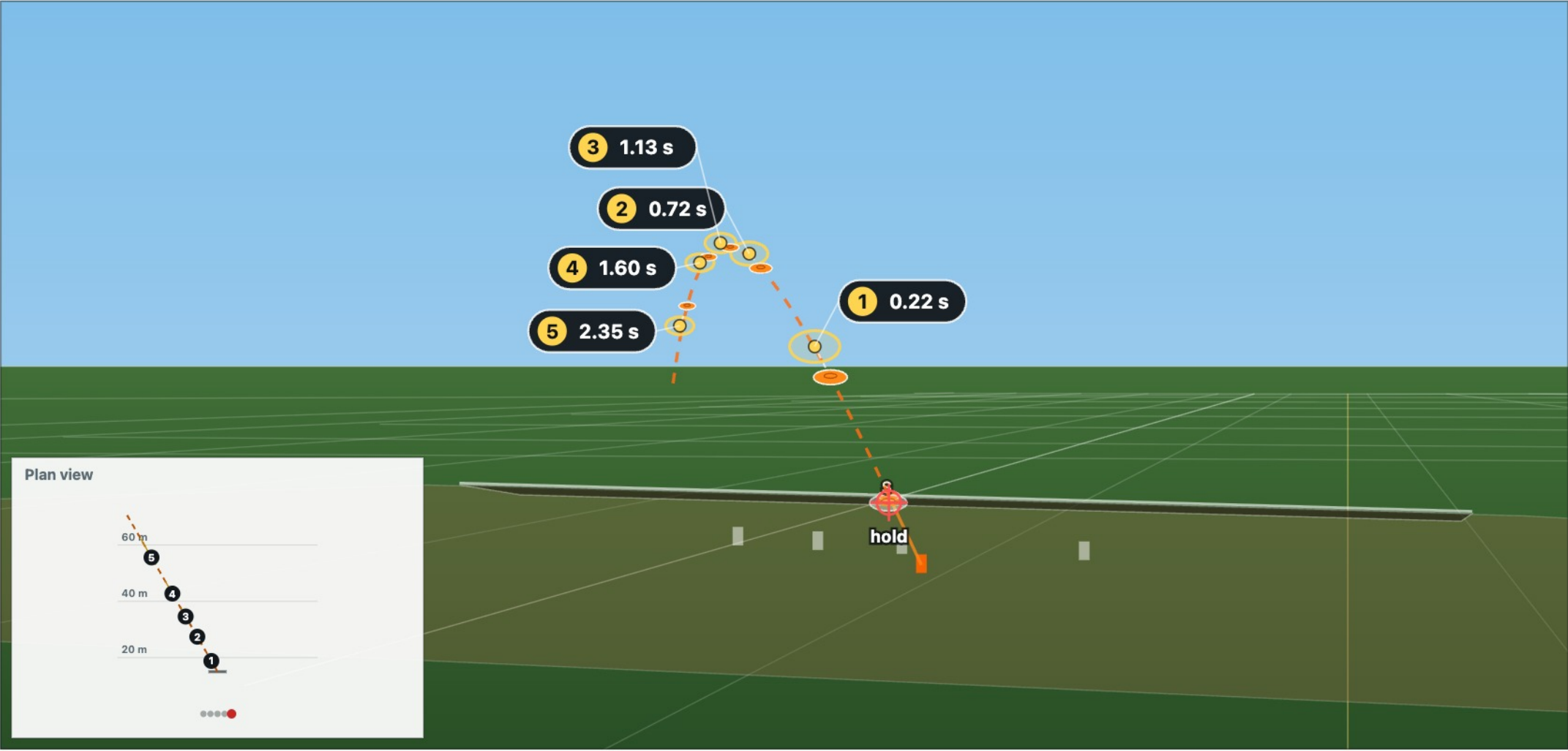
1 Release 0.22 s after release 56 cm ≈ 5.1 clays 32 cm right · 45 cm up +0.81° lateral · +1.14° vertical clay at 23 m · height 1.1 m · real separation 3.2 m first glimpse · within reach	2 Early 0.51 s after release 47 cm ≈ 4.3 clays 35 cm right · 32 cm up +0.64° lateral · +0.57° vertical clay at 32 m · height 2.1 m · real separation 4.8 m quick shot · within reach	3 Good window 0.75 s after release 40 cm ≈ 3.7 clays 38 cm right · 12 cm up +0.56° lateral · +0.18° vertical clay at 39 m · height 2.7 m · real separation 6.5 m the comfortable shot · at the limit	4 At the limit 1.02 s after release 47 cm ≈ 4.3 clays 42 cm right · 21 cm down +0.51° lateral · -0.25° vertical clay at 48 m · height 3.0 m · real separation 8.7 m last useful shot · at the limit	5 You are late 1.63 s after release 140 cm ≈ 12.7 clays 46 cm right · 132 cm down +0.39° lateral · -1.13° vertical clay at 67 m · height 2.6 m · real separation 13.3 m no longer reaching well · beyond useful range
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Station 5 · Machine 4

angle -30° · height 3.0 m at 10 m · fall 65 m · 76 km/h · elevation 18.6°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 1.58x over the base field of 48° vertical · view turned -4° horizontally and +5° vertically



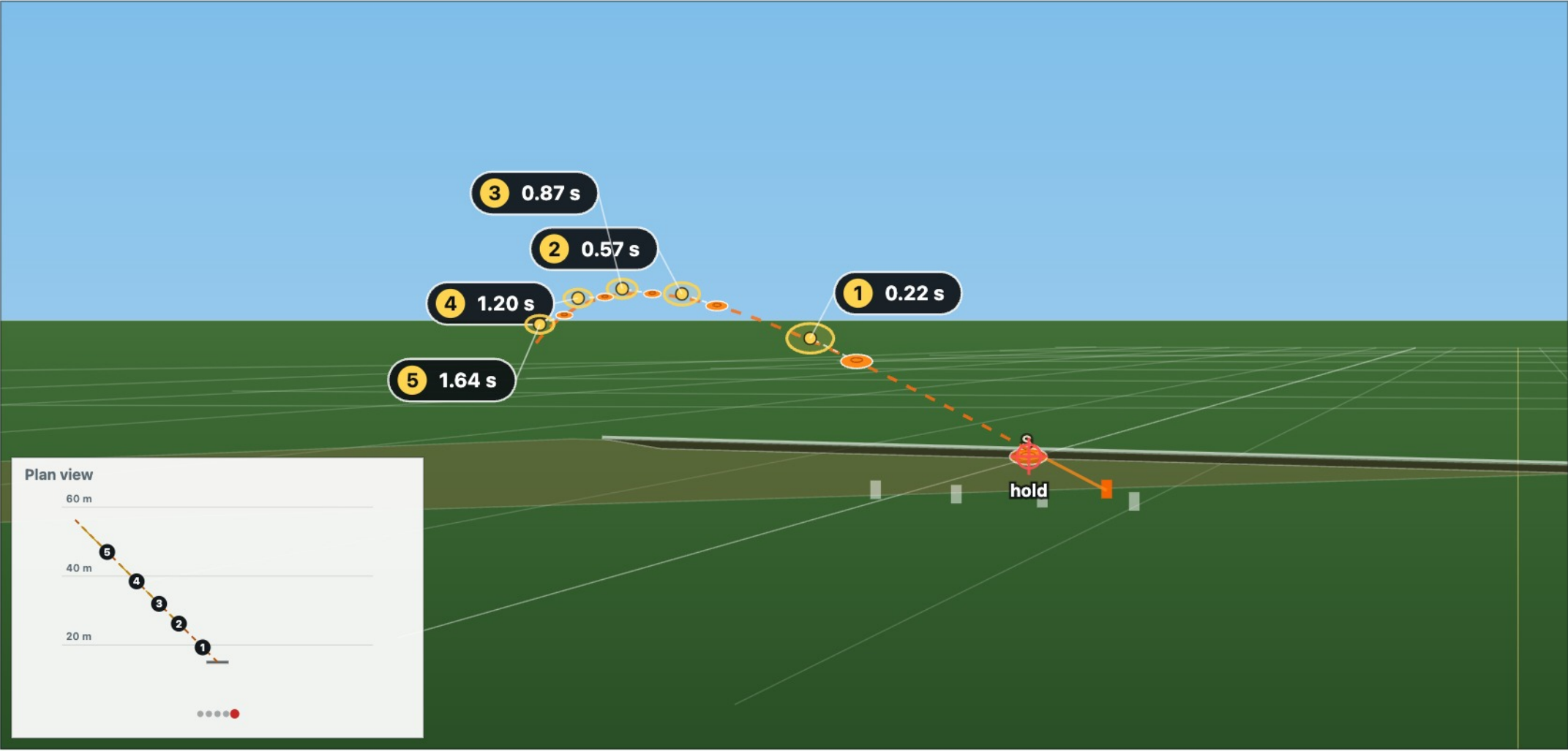
1 Release 0.22 s after release 54 cm ≈ 4.9 clays 26 cm left · 47 cm up -0.73° lateral · +1.34° vertical clay at 20 m · height 1.5 m · real separation 1.7 m first glimpse · within reach	2 Early 0.72 s after release 39 cm ≈ 3.5 clays 28 cm left · 27 cm up -0.54° lateral · +0.51° vertical clay at 30 m · height 3.9 m · real separation 2.7 m quick shot · within reach	3 Good window 1.13 s after release 30 cm ≈ 2.8 clays 30 cm left · 2 cm down -0.45° lateral · -0.03° vertical clay at 38 m · height 5.1 m · real separation 3.7 m the comfortable shot · at the limit	4 At the limit 1.60 s after release 61 cm ≈ 5.6 clays 33 cm left · 51 cm down -0.40° lateral · -0.62° vertical clay at 48 m · height 5.6 m · real separation 5.0 m last useful shot · at the limit	5 You are late 2.35 s after release 182 cm ≈ 16.6 clays 39 cm left · 178 cm down -0.35° lateral · -1.63° vertical clay at 62 m · height 4.5 m · real separation 7.7 m no longer reaching well · beyond useful range
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Station 5 • Machine 5

angle -45° • height 1.5 m at 10 m • fall 60 m • 101 km/h • elevation 9.1°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s • rib 0% (0.00°) • wind 0.0 m/s • delay 0.03 s • pattern 1 3/4 • hold neutral
framing 1.58x over the base field of 48° vertical • view turned -10° horizontally and +3° vertically



1 Release 0.22 s after release 91 cm ≈ 8.3 clays 83 cm left • 38 cm up -2.21° lateral • +1.01° vertical clay at 21 m • height 1.0 m • real separation 2.5 m first glimpse • within reach	2 Early 0.57 s after release 93 cm ≈ 8.5 clays 91 cm left • 22 cm up -1.69° lateral • +0.40° vertical clay at 31 m • height 2.0 m • real separation 4.0 m quick shot • within reach	3 Good window 0.87 s after release 99 cm ≈ 9.0 clays 99 cm left • 1 cm down -1.46° lateral • -0.02° vertical clay at 39 m • height 2.5 m • real separation 5.4 m the comfortable shot • at the limit	4 At the limit 1.20 s after release 116 cm ≈ 10.6 clays 109 cm left • 41 cm down -1.30° lateral • -0.49° vertical clay at 48 m • height 2.5 m • real separation 7.4 m last useful shot • at the limit	5 You are late 1.64 s after release 175 cm ≈ 16.0 clays 126 cm left • 122 cm down -1.20° lateral • -1.17° vertical clay at 60 m • height 1.9 m • real separation 10.7 m no longer reaching well • beyond useful range
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Read it before applying anything from this notebook on a shooting ground

This is an estimate, not a measurement

The trajectories, the times and the leads come from a simplified model that solves a visual parabola from the height and the range of the clay, with reduced gravity and a common visual release. It does not measure a real trap, it does not reproduce the full aerodynamics of the disc, and it does not replace a ballistic table for the cartridge you use.

The scheme tables reproduce values published by the federations, but this document does not certify any installation or any trajectory. In competition the authorised officials always decide.

The responsibility for the shot is yours

Use these figures as a starting point for observing and comparing, never as an instruction to follow blindly. Check your own numbers on the ground, with your gun, your cartridge and the conditions of the day, and correct whatever needs correcting.

This material does not replace in-person training, the rules of the ground, the help of an instructor or a gunsmith, or the legal requirements in force. Safety and shooting decisions are the responsibility of whoever pulls the trigger.

Before taking it to the ground

- Check that the scheme on the cover really is the one that is going to be thrown. Ask whoever is in charge instead of assuming it.
- The leads depend on the cartridge and the rib you set when generating the notebook. With different equipment, the figures change.
- Treat every number as a hypothesis you are going to test by shooting, not as the result you ought to get.

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Where everything is

Full book, simulator and source code:

www.robertocanales.com/proyectos#plato

rcanalescoder.github.io/Plato

rcanalesmora@gmail.com

The project is in beta

The book and the simulator are still under review and are shared so that anyone can try them, find errors and help improve them. If you find something that does not match your experience on the ground, the rules that apply to you, or the figures you measure yourself, write in: that kind of report is what makes the model better.